The Journal of Child Psychology and Psychiatry

Journal of Child Psychology and Psychiatry 63:4 (2022), pp 381–396 doi:10.1111/jcpp.13534

Annual Research Review: Shifting from 'normal science' to neurodiversity in autism science

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Since its initial description, the concept of autism has been firmly rooted within the conventional medical paradigm of child psychiatry. Increasingly, there have been calls from the autistic community and, more recently, nonautistic researchers, to rethink the way in which autism science is framed and conducted. Neurodiversity, where autism is seen as one form of variation within a diversity of minds, has been proposed as a potential alternative paradigm. In this review, we concentrate on three major challenges to the conventional medical paradigm – an overfocus on deficits, an emphasis on the individual as opposed to their broader context and a narrowness of perspective – each of which necessarily constrains what we can know about autism and how we are able to know it. We then outline the ways in which fundamental elements of the neurodiversity paradigm can potentially help researchers respond to the medical model's limitations. We conclude by considering the implications of a shift towards the neurodiversity paradigm for autism science. **Keywords:** Autism; ethics; medical model; neurodiversity; social model of disability.

Introduction

Science is not static. As Thomas Kuhn (1962) explained, science progresses through a series of phases from what Kuhn called 'normal science' – the accepted orthodoxy of the moment – to periods of crisis, when scientists begin to contest the hitherto-accepted paradigm itself. This period ends, ultimately, in a shift from one paradigm to another. In the field of autism science, the conventional medical paradigm is – and has long been – the accepted orthodoxy in this field, conceptualising autism in terms of biologically derived functional deficits, and thus placing limits or boundaries on what we can know about autism and how we are able to know it (Kuhn, 1962). The vast majority of autism researchers have been trained to understand autism as a disorder of brain development, an undesirable deviation from the norm.

There have been 'rumblings' in autism science, however, of the sort that Kuhn described. In a context of social change, with many challenges to established power structures, autistic advocates and autism scientists have increasingly called to replace the conventional medical paradigm and consider autism instead through the lens of *neurodiversity*, where autism is seen as one form of variation within a diversity of minds (Singer, 1998; Walker & Raymaker, 2020). These calls, stemming originally from autistic[1] activists (Pripas-Kapit, 2020; Sinclair, 1993) have increasingly found at least partial adherents from within autism science (Baron-Cohen, 2000; Gernsbacher, 2007; Happé & Frith, 2020; Mottron, 2011; Nicolaidis, 2012), suggesting that researchers could be on the brink of thinking about autism in a fundamentally different way. Doing so could radically change how we approach knowledge construction within autism science and the way that we support autistic people and their families in our practice.

In what follows below, we proceed in two major sections. First, we outline the major ways in which the conventional medical paradigm is being called into question. Second, we outline the fundamental elements of the alternative view, the neurodiversity paradigm. We briefly trace its history, describe its core tenets and ask whether the neurodiversity paradigm could potentially overcome the challenges faced by its medical counterpart. We then conclude by considering the implications of a shift towards the neurodiversity paradigm for autism science.

The conventional medical paradigm

The conventional medical paradigm, also known as the medical model of disability (Llewellyn & Hogan, 2000; Marks, 1997), approaches autism as a disability primarily rooted *within* individuals. Within the medical paradigm, disability is seen to arise as a direct consequence of a person's biological make-up and functioning. Specifically, disability is defined as 'any restriction or lack (resulting from an impairment) of ability to perform an activity in the manner or within the range considered normal for a human being' (World Health Organisation, 1980, p. 143). As demonstrated in this definition, the conventional medical paradigm implicitly assumes the existence of a typical or 'normal' level of ability that is held up as the ideal 'state of health'; those who enjoy this state are taken as the normative standard and any

Conflict of interest statement: No conflicts declared.

The Journal of Child Psychology and Psychiatry

儿童心理学与精神病学杂志 63:4 (2022年), pp 381–396 doi:10.1111/jcpp.13534

年度研究综述：从'常规科学'转向神经多样性范式在孤独谱系科学中的应用

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自其最初被描述以来，孤独谱系的概念一直牢固地植根于儿童精神病学的传统医学范式之中。越来越多地，孤独谱系社群以及近来非孤独谱系研究者呼吁重新思考孤独谱系科学的构建与开展方式。神经多样性——将孤独谱系视为多样心智中的一种变异形式——已被提出作为一种潜在的替代范式。在本综述中，我们聚焦于传统医学范式的三大挑战：过度聚焦于缺陷、强调个体而非其更广泛的环境以及视角的狭隘性，每一项都必然限制了我们对孤独谱系的认知以及认知的方式。随后，我们概述了神经多样性范式的基本要素如何可能帮助研究者应对医学模式的局限。最后，我们探讨了转向神经多样性范式对孤独谱系科学的意义。关键词：孤独谱系；伦理；医学模式；神经多样性；残障社会模式。

引言

科学并非静止不变。正如Thomas Kuhn（1962）所阐述的，科学经历一系列阶段而发展，从Kuhn所称的'常规科学'——即当下被接受的正统观念——到危机时期，此时科学家开始质疑迄今为止被接受的范式本身。这一阶段最终以从一种范式转向另一种范式而结束。在孤独谱系科学领域，传统的医学范式一直是该领域被接受的正统观念，它将孤独谱系概念化为源于生物学的功能性缺陷，从而对我们能了解孤独谱系的范畴以及我们了解它的方式设定了限制或边界（Kuhn, 1962）。绝大多数孤独谱系研究者被训练成将孤独谱系理解为一种脑发育障碍，一种偏离常态的不理想偏差。

然而，孤独谱系科学中已出现 Kuhn 所描述的那种'骚动'。在社会变革的背景下，随着对既有权力结构的诸多挑战，孤独谱系倡导者和孤独谱系科学家越来越多地呼吁取代传统的医学范式，转而通过神经多样性的视角来看待孤独谱系，即将其视为多样心智中的一种变异形式（Singer, 1998; Walker & Raymaker, 2020）。这些呼吁最初源于孤独谱系[1] 活动家（Pripas-Kapit, 2020; Sinclair, 1993），如今越来越多地获得了孤独谱系科学界内至少部分人士的认同（Baron-Cohen, 2000; Gernsbacher, 2007; Happ ́e & Frith, 2020; Mottron, 2011; Nicolaidis, 2012），这表明研究人员可能正处于从根本上以不同方式思考孤独谱系的边缘。这样做可能会彻底改变我们在孤独谱系科学中构建知识的方式，以及我们在实践中支持孤独谱系群体及其家庭的方式。

在接下来的内容中，我们将分两个主要部分展开。首先，我们概述传统医学范式被质疑的主要方面。其次，我们概述替代性观点——神经多样性范式的基本要素。我们简要追溯其历史，描述其核心原则，并探讨神经多样性范式是否有可能克服其医学对应范式所面临的挑战。最后，我们通过思考向神经多样性范式转变对孤独谱系科学的影响来得出结论。

传统的医学范式

传统的医学范式，也称为残疾医学模型（Llewellyn & Hogan, 2000; Marks, 1997），将孤独谱系视为一种主要根植于个体内部的残疾。在医学范式内，残疾被视为个人生物构成与功能的直接后果。具体而言，残疾被定义为"（由损伤导致的）进行某项活动的能力受限或缺失，其方式或范围被认为不符合人类正常标准"（世界卫生组织，1980年，第143页）。正如该定义所示，传统的医学范式隐含地假定存在一种典型或'正常'的能力水平，并将其视为理想的'健康状态'；享有这种状态的人被视为规范性标准，而任何

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other is unfavourably compared with it (Akhtar & Jaswal, 2013; Medin, Bennis, & Chandler, 2010).

In line with this perception of disability as an undesirable feature of the individual, treatment of disability under the medical paradigm typically aims to bring an individual's abilities into line with the accepted norm. Treatments and/or interventions are therefore applied to the disabled person, with the goal of altering the individual's impairment/s in order to remediate or eliminate disability and enhance functioning.

Widespread understandings of autism have firm roots within this medical paradigm (Evans, 2013; Fletcher-Watson & Happé, 2019; Silberman, 2015). The first published descriptions of the constellation of traits we now call autism were authored by a psychiatrist (Kanner, 1943) and a paediatrician (Asperger, 1944) working within the conventional medical paradigm of their time. Today, autism is listed in both the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) and the International Classification of Diseases (ICD-11) as 'Autism Spectrum Disorder', a neurodevelopmental disorder. In both manuals, Autism Spectrum Disorder is described as a series of 'persistent deficits' demonstrated by autistic children, young people and adults, involving deficits in social communication and interaction, and restricted, repetitive and inflexible patterns of behaviour, interests or activities.

This manner of characterising autism focuses solely on the autistic person and their perceived 'deficits'. Consequently, one key aim of autism science has been to identify the specific neurodevelopmental mechanisms – at the genetic, neurobiological and cognitive levels – that might explain the behavioural manifestations of autism. The implicit argument is that such scientific progress is an essential precondition for any further translational efforts to inform treatments and interventions for autistic children, helping to 'guide brain and behavioural development back toward a normal pathway' (Dawson, 2008, p. 776).

This approach has yielded a number of scientific breakthroughs that have dramatically advanced our understanding of autism (see Happé & Frith, 2020, for review). But challenges have been increasingly loud of late, in part due to the rise in autistic self-advocacy and the neurodiversity movement, and in part due the relative absence of 'hard facts', or universal scientific truths about autism, at any level of explanation (see Verhoeff, 2015, for discussion). As Professor Sir Michael Rutter put it: 'it seems decidedly odd that after more than half a century of both research and clinical experience with Autism Spectrum Disorders, there continue to be arguments on the nature of autism" (2014, p. 55). For some, it is time to consider whether we should be approaching autism science and practice in a radically different way.

A focus on deficits

The conventional medical approach searches for impairments and functional deficits in autistic people and often has the unintended consequence of drawing attention away from the particular strengths of autistic people and focusing entirely on limitations, whether perceived or real. It detracts, in other words, from an account of what autistic people *can* do and stresses what they *cannot*. It does so despite longstanding recognition that autistic people might excel at particular activities in individual instances (Frith & Happé, 1994; Hermelin & Frith, 1971; Hermelin & O'Connor, 1975; Mottron & Belleville, 1993; Shah & Frith, 1983). An increasing number of studies show that while autistic people outperform nonautistic people on a variety of tasks (e.g. Muth, Honekopp, & Falter, 2014; Remington & Fairnie, 2017; Samson, Mottron, Soulieres, & Zeffiro, 2012), these strengths are very rarely considered in the profile of autism (American Psychiatric Association, 2013). When mentioned at all, these tend to be referred to as 'islets of abilities' among a sea of deficits (Shah & Frith, 1983). The adoption of a predominantly deficit-focused view therefore tends to present autism as a lack or absence of something that someone ought to have, an undesirable individual experience (Dinishak, 2016; Robertson, 2010).

The problem goes deeper than this, however. In a number of cases, conventional autism research describes ways in which autistic people *outperform* nonautistic people in scientific tasks yet interpret those achievements as somehow revealing a problem. That is, data that in fact reveal strengths in autistic people are paradoxically – and bizarrely – interpreted in a negative way, as a consequence of a 'deficit' or 'impairment' (Dawson & Mottron, 2011; Dinishak, 2016; Gernsbacher, Dawson, & Mottron, 2006; Robertson, 2010). One of the authors of this review has previously fallen foul of this deficit-focused bias herself. She examined the extent to which autistic children were susceptible to perceptual aftereffects – a change in subjective perceptual experience following prolonged exposure to a stimulus (Webster, 2011). In that study, she initially reported that autistic children showed a significantly *reduced* perceptual after effect for faces compared with nonautistic children of similar age and ability, as if that was a sign of a functional deficit (Pellicano, Jeffery, Burr, & Rhodes, 2007). Yet it soon became clear the same data could be read in almost exactly the opposite way: autistic children's face recognition following adaptation was actually *more accurate* than that of nonautistic children. Whereas nonautistic children were led astray by their preconceptions, the autistic children's perception corresponded more to physical reality than to expectations (i.e. it was more veridical; Pellicano & Burr, 2012).

Other authors make similar observations with regard to a range of autistic social and nonsocial

其他则被与之不利地比较 (Akhtar & Jaswal, 2013; Medin, Bennis, & Chandler, 2010)。

与这种将残障视为个体不良特征的认知相一致，医学范式下的残障治疗通常旨在使个体的能力符合公认的规范。因此，治疗和/或干预措施被应用于残障人士，目标是改变个体的损伤，以纠正或消除残障并提升功能水平。

对孤独谱系的普遍理解深深植根于这一医学范式（Evans, 2013; Fletcher-Watson & Happ´e, 2019; Silberman, 2015）。如今我们称之为孤独谱系的特质组合，其首次公开发表的描述是由一位精神病学家（Kanner, 1943）和一位儿科医生（Asperger, 1944）在他们所处时代的传统医学范式下完成的。如今，孤独谱系在《精神障碍诊断与统计手册》（DSM-5）和《国际疾病分类》（ICD-11）中均被列为'孤独谱系障碍'，一种神经发育障碍。在两部手册中，孤独谱系障碍都被描述为一系列由孤独谱系儿童、青少年和成人所表现的'持续性缺陷'，涉及社交沟通与互动方面的缺陷，以及行为、兴趣或活动的受限、重复和僵化模式。

这种描述孤独谱系的方式仅聚焦于孤独谱系个体及其被认为存在的'缺陷'。因此，孤独谱系科学的一个关键目标便是识别特定的神经发育机制——在遗传、神经生物学和认知层面——这些机制或许能解释孤独谱系的行为表现。隐含的论点是，此类科学进展是任何进一步转化努力的必要前提，旨在为孤独谱系儿童的治疗和干预提供信息，帮助'引导大脑和行为发展回归正常轨道'（Dawson, 2008, p. 776）。

这种方法已催生了许多科学突破，极大地推进了我们对孤独谱系的理解（综述参见 Happ´e & Frith, 2020）。但近来质疑声日益高涨，部分原因是孤独谱系群体自我倡导的兴起和神经多样性运动，部分原因则是在任何解释层面上，关于孤独谱系的"硬事实"或普遍科学真理相对匮乏（讨论参见 Verhoeff, 2015）。正如 Michael Rutter 爵士教授所言："在对孤独谱系障碍进行了超过半个世纪的研究和临床实践之后，关于孤独谱系本质的争论仍在持续，这着实令人感到奇怪"（2014, p. 55）。对一些人而言，现在是时候考虑是否应以一种截然不同的方式来对待孤独谱系科学和实践了。

聚焦于缺陷

传统的医学方法致力于寻找孤独谱系群体的损伤和功能缺陷，往往无意中将注意力从孤独谱系群体的特定优势上转移开，而完全聚焦于其局限——无论是感知到的还是真实的。换言之，它削弱了对孤独谱系群体所能之事的描述，并强调其所不能之事。尽管长期以来人们认识到，在个别情况下，孤独谱系群体可能在特定活动中表现卓越（Frith & Happ´e, 1994; Hermelin & Frith, 1971; Hermelin & O'Connor, 1975; Mottron & Belleville, 1993; Shah & Frith, 1983），但这种方法依然如此行事。越来越多的研究表明，虽然孤独谱系群体在多种任务上表现优于非孤独谱系群体（例如 Muth, Honekopp, & Falter, 2014; Remington & Fairnie, 2017; Samson, Mottron, Soulieres, & Zeffiro, 2012），但这些优势极少被纳入孤独谱系的特征描述中（美国精神医学学会, 2013）。即便提及，这些优势也往往被称为缺陷海洋中的"能力孤岛"（Shah & Frith, 1983）。因此，采用一种主要关注缺陷的观点，倾向于将孤独谱系呈现为某人本应拥有之物的缺失或匮乏，一种不受欢迎的个人体验（Dinishak, 2016; Robertson, 2010）。

然而，问题比这更深层。在许多案例中，传统的孤独谱系研究描述了孤独谱系群体在科学任务中表现优于非孤独谱系群体的方式，却将这些成就解读为某种程度上揭示了一个问题。也就是说，那些事实上揭示孤独谱系群体优势的数据，被悖论性地——且奇怪地——以负面方式解读，归因于某种'缺陷'或'损伤'（Dawson & Mottron, 2011; Dinishak, 2016; Gernsbacher, Dawson, & Mottron, 2006; Robertson, 2010）。本综述的一位作者此前也曾陷入这种聚焦缺陷的偏见。她研究了孤独谱系儿童对知觉后效的敏感性程度——即长时间暴露于某种刺激后主观知觉体验的变化（Webster, 2011）。在那项研究中，她最初报告称，与年龄和能力相似的非孤独谱系儿童相比，孤独谱系儿童对面孔的知觉后效显著减弱，仿佛这是功能缺陷的标志（Pellicano, Jeffery, Burr, & Rhodes, 2007）。然而很快便发现，同样的数据几乎可以完全以相反的方式解读：适应后，孤独谱系儿童的面孔识别实际上比非孤独谱系儿童更准确。非孤独谱系儿童被他们的先入之见误导，而孤独谱系儿童的知觉更符合物理现实而非预期（即更真实；Pellicano & Burr, 2012）。

其他作者在一系列孤独谱系社交与非社交方面也做出了类似的观察。

'deficits' which could equally be seen as 'strengths' (Bertilsdotter Rosqvist & Jackson-Perry, 2020; Dawson, Soulieres, Gernsbacher, & Mottron, 2007; Gernsbacher et al., 2006; Mottron, 2011; Robertson, 2010). Even when autistic people show enhanced performance on visuo-perceptual tasks, like the Block Design task from the Wechsler Scales of Intelligence (Wechsler, 2008, 2014) or on the Embedded Figures Test (Witkin, Oltman, Raskin, & Karp, 1971), such strengths are invariably characterised as a by-product of a deficit in higher-order cognition (e.g. Frith & Happé's 'weak central coherence', 1994). As Dawson and Mottron (2011) write:

> Autistics, like non-autistics, have genuine difficulties in many areas, and like non-autistics, require assistance in areas where their performance is weak… But autistics uniquely are seen as pathological when displaying significant or dramatic strengths, creating for autistics a nearly insurmountable disadvantage or disability not faced by non-autistics (p. 34)

This tendency to interpret autistic performance negatively is seen further in the research literature on autistic intelligence, which demonstrates that it is often the research design itself that is the cause of the issue. For many years, researchers interpreted autistic people's low scores on, or noncompletion of, standard intelligence tests (e.g. Wechsler Scales of Intelligence) simply as confirmation of intellectual disability. In fact, however, once strength-informed intelligence tests (e.g. Raven's Progressive Matrices) were substituted for the conventional ones, those deficits disappeared (Courchesne, Girard, Jacques, & Soulières, 2015; Courchesne, Meilleur, Poulin-Lord, Dawson, & Soulières, 2019; Dawson et al., 2007).

Such negative interpretations and the research design that reinforces them have consequences beyond research itself. Autistic scientist, Michelle Dawson, has long argued that the habit of casting autistic people as 'less than' has resulted in autistic people being subject to medical and other interventions that are not as fully supported by evidence as they should be. This is particularly the case with one dominant intervention, Applied Behavioural Analysis (Dawson, 2004). Even as recently as 2019, autistic people have been subjected to 'aversive' treatments in behavioural intervention research (Verriden & Roscoe, 2019), including electric shocks as punishment at the highly controversial Judge Rotenberg Educational Center of Canton, Massachusetts, which is still open for business despite having been condemned for torture by the United Nations Special Rapporteur on Torture (Neumeier & Brown, 2020). The persistent focus on deficits serves to support these dehumanising attitudes; seeing autistic people as 'less than human' (Goffman, 1990; see also Cage, di Monaco, & Newell, 2019) legitimises the use of electric shock in this instance.

Dawson frames these 'unacceptably low standards' (Cowen & Dawson, 2018, para 146) as human rights issues and calls for researchers and practitioners to ensure basic standards in research and practice are extended to autistic people (see Dawson & Fletcher-Watson, 2021). The relative failure of autism researchers to report potential conflicts of interest in research (Bottema-Beutel, Crowley, Sandbank, & Woynaroski, 2021b) and adverse events or potential 'harms' in nonpharmacological interventions for young autistic children (Bottema-Beutel, Crowley, Sandbank, & Woynaroski, 2021a; Bottema-Beutel, Kapp, Lester, Sasson, & Hand, 2021) is consistent with this view (Davis, den Houting, Nordahl-Hansen, & Fletcher-Watson, in press).

Aside from concerns regarding questionable standards in autism intervention, the conceptualisation of autism as a series of deficits and the goal of intervention thus to 'make people less autistic' is morally troubling in itself (Ne'eman, 2021; see also Callanan & Waxman, 2013). Autistic people and their families are regularly presented with the message that being autistic is a tragic fate, with an autism diagnosis presumed to prompt grief and mourning. The language of deficits dominates discussion of autism both in scientific writings and in the popular press (Bottema-Beutel et al., 2021; Kenny et al., 2016). Researchers and others use terms like 'disorder', 'deficit', 'impairment' and 'dysfunction' to describe most deviations from the norm in autistic behaviour, cognition or neurobiology. Researchers commonly use person-first language ('child with autism'), as if autism can be separated from the person, even in the knowledge that many autistic people eschew this terminology (Kenny et al., 2016; Sinclair, 2012) and that it is more likely to contribute to stigma (Bottema-Beutel et al., 2021; Gernsbacher, 2017). Researchers also routinely describe children as 'high-' or 'low-functioning' (Alvares et al., 2020; Kenny et al., 2016), as being 'at risk' of developing autism (Jones, Gliga, Bedford, Charman, & Johnson, 2014; Kolevzon, Gross, & Reichenberg, 2007; Modabbernia, Velthorst, & Reichenberg, 2017; Yudell, Tabor, Dawson, Rossi, & Newschaffer, 2013), and as having the potential to achieve an 'optimal outcome' (Fein et al., 2013; Orinstein et al., 2014; Sutera et al., 2007) – where optimality is equated with no longer meeting diagnostic criteria for autism. Frequently, policy makers and others discuss autistic people in terms of 'burden', in relation to both the economic costs associated with autism (e.g. Knapp, Romeo, & Beecham, 2009; Leigh & Du, 2015) and the experiences of parents and others who care for autistic people (e.g. Marsack-Topolewski, Samuel, & Tarraf, 2021; Picardi et al., 2018).

In the face of this constant negative messaging, especially combined with more generalised discrimination against disabled people, it is unsurprising that considerable stigma is associated with the

这些'缺陷'同样可以被视为'优势'（Bertilsdotter Rosqvist & Jackson-Perry, 2020; Dawson, Soulieres, Gernsbacher, & Mottron, 2007; Gernsbacher et al., 2006; Mottron, 2011; Robertson, 2010）。即使孤独谱系群体在视觉感知任务上表现出增强的能力，例如韦氏智力量表中的积木设计任务（Wechsler, 2008, 2014）或镶嵌图形测验（Witkin, Oltman, Raskin, & Karp, 1971），此类优势也总是被描述为高阶认知缺陷的副产品（例如Frith & Happ´e 的'弱中央统合'，1994）。正如Dawson 和 Mottron（2011）写道：

> 孤独谱系群体与非孤独谱系群体一样，在许多领域确实存在困难，并且与非孤独谱系群体一样，在其表现薄弱的领域需要协助。然而，孤独谱系群体独特之处在于，当他们展现出显著或惊人的优势时，却被视为病态，这为孤独谱系群体创造了几乎无法克服的劣势或障碍，而这是非孤独谱系群体所不曾面对的（第34页）。

这种对孤独谱系表现进行负面解读的倾向，在关于孤独谱系智力的研究文献中进一步显现，文献表明，问题往往源于研究设计本身。多年来，研究人员将孤独谱系群体在标准智力测试（例如韦氏智力量表）上的低分或未能完成测试，简单地解读为智力缺陷的确认。然而事实上，一旦采用基于优势的智力测试（例如瑞文渐进矩阵）替代传统测试，那些缺陷便消失了（Courchesne, Girard, Jacques, & Souli`eres, 2015; Courchesne, Meilleur, Poulin-Lord, Dawson, & Souli`eres, 2019; Dawson et al., 2007）。

这种负面解读以及强化它们的研究设计，其影响已超出研究本身。孤独谱系科学家Michelle Dawson长期以来一直主张，将孤独谱系群体视为'次等'的习惯，导致孤独谱系群体承受了医学及其他干预措施，而这些措施的证据支持程度并未达到应有的充分水平。这在一种主导性干预方法——应用行为分析（Dawson, 2004）中尤为明显。即便在最近的2019年，孤独谱系群体仍在行为干预研究中遭受'厌恶式'治疗（Verriden & Roscoe, 2019），包括在备受争议的马萨诸塞州坎顿Judge Rotenberg教育中心使用电击作为惩罚手段。该中心尽管已被联合国酷刑问题特别报告员谴责为酷刑（Neumeier & Brown, 2020），却仍在运营。对缺陷的持续关注助长了这些去人性化的态度；将孤独谱系群体视为'低于人类'（Goffman, 1990；另见Cage、di Monaco与Newell, 2019）在此例中为使用电击提供了正当性依据。

Dawson将这些'不可接受的低标准'（Cowen与Dawson, 2018, 第146段）框定为人权问题，并呼吁研究者与实践者确保研究与实践中基本标准能延伸至孤独谱系群体（参见Dawson与Fletcher-Watson, 2021）。孤独谱系研究者相对未能报告研究中的潜在利益冲突（Bottema-Beutel、Crowley、Sandbank与Woynaroski, 2021b），以及针对孤独谱系幼儿的非药物干预中的不良事件或潜在'伤害'（Bottema-Beutel、Crowley、Sandbank与Woynaroski, 2021a; Bottema-Beutel、Kapp、Lester、Sasson与Hand, 2021），这与这一观点是一致的（Davis、den Houting、Nordahl-Hansen与Fletcher-Watson, 即将出版）。

除了对孤独谱系干预中可疑标准的担忧外，将孤独谱系概念化为一系列缺陷，并将干预目标设定为'让人们不那么孤独谱系'，这本身在道德上就令人困扰（Ne'eman, 2021；另见Callanan与Waxman, 2013）。孤独谱系群体及其家庭经常接收到这样的信息：成为孤独谱系是一种悲惨的命运，孤独谱系诊断被认为会引发悲伤和哀悼。缺陷的语言主导着对孤独谱系的讨论，无论是在科学著作还是在大众媒体中（Bottema-Beutel等, 2021; Kenny等, 2016）。研究人员和其他人使用诸如'障碍'、'缺陷'、'损伤'和'功能障碍'等术语来描述孤独谱系行为、认知或神经生物学中大多数偏离常态的情况。研究人员普遍使用以人为本语言（'孤独谱系儿童'），仿佛孤独谱系可以与个人分离，尽管他们知道许多孤独谱系群体摒弃这种术语（Kenny等, 2016；Sinclair, 2012），并且它更可能助长污名（Bottema-Beutel等, 2021; Gernsbacher, 2017）。研究人员还例行公事地将儿童描述为'高功能'或'低功能'（Alvares等, 2020; Kenny等, 2016），描述为有发展孤独谱系的'风险'（Jones、Gliga、Bedford、Charman与Johnson, 2014；Kolevzon、Gross与Reichenberg, 2007; Modabbernia、Velthorst与Reichenberg, 2017; Yudell、Tabor、Dawson、Rossi与Newschaffer, 2013），以及描述为有潜力实现'最佳结果'（Fein等, 2013；Orinstein等, 2014；Sutera等, 2007）——其中最佳性等同于不再符合孤独谱系的诊断标准。政策制定者和其他人经常以'负担'来讨论孤独谱系群体，这既涉及与孤独谱系相关的经济成本（例如Knapp、Romeo与Beecham, 2009；Leigh与Du, 2015），也涉及父母和其他照顾孤独谱系群体的经历（例如Marsack-Topolewski、Samuel与Tarraf, 2021; Picardi等, 2018）。

面对这种持续不断的负面信息传递，尤其是与针对残障群体的更普遍歧视相结合时，与孤独谱系标签相关的严重污名也就不足为奇了。

autism label. Studies with young autistic people in particular have found that they often construe themselves as 'different' to neurotypical people in a negative way (Cribb, Kenny, & Pellicano, 2019; Humphrey & Lewis, 2008; Shattuck et al., 2014; Williams, Gleeson, & Jones, 2019). They label themselves as a 'freak', 'mentally disabled' and as 'having a bad brain' (Humphrey & Lewis, 2008), and often just want to be 'as normal as possible', actively hiding their autistic-ness (Cribb et al., 2019). These negative self-appraisals are not unexpected given that young autistic people are more likely to experience social exclusion and bullying (Humphrey & Hebron, 2014; Maiano, Normand, Salvas, Moullec, & Aime, 2016; Schroeder, Cappadocia, Bebko, Pepler, & Weiss, 2014). But they can have damaging effects on their mental health (Cage, Di Monaco, & Newell, 2018; Cooper, Smith, & Russell, 2017; Hedley & Young, 2006), where a disabled/autistic person may come to believe that they are less worthy and act accordingly (Becker, 1963; Milton, 2012). Actively hiding one's autistic differences through masking or camouflaging (see Pearson & Rose, 2021, for discussion) can also come at serious personal cost, including stress, anxiety and negative self-perceptions (Hull et al., 2017) and burnout (Higgins, Arnold, Weise, Pellicano, & Trollor, 2021; Raymaker et al., 2020) in autistic adults.

The medical model is essentially individualist

The prevalence of deficit-based thinking has the further consequence of focusing attention directly on the individual and away from social and environmental factors that might in fact play a significant role in shaping autistic lives (Engel, 1977). In the conventional medical view, autism and its associated disabilities are seen as something inherent to the individual. Biomedical research thus tends to explain an autistic person's difficulties not with reference to the context in which the difficulty occurs – home, school, work or broader community – but rather as a characteristic of the individual themselves. Within autism science, this tendency has been further enhanced by the well-established portrayal of the autistic person as being on their own in the world (Broderick & Ne'eman, 2008). Given that autism was originally seen to manifest as withdrawal from the world (Kanner, 1943), it made even more intuitive sense to early researchers to propose that the features that determine the nature of the autistic experience lay with the individual themselves.

Taking this individualistic starting point suggests that the 'fault' for difficulties in life resides with the individual themselves, thus the burden of 'correcting' perceived difficulties lies there too. The autistic person is perceived to be in some way 'flawed' or 'defective', with individual treatment required to remediate these shortcomings. In autism research and practice, it is clearly evident that autistic people are by and large expected to 'overcome' their impairment/s in order to achieve a typical level of ability. Treatments and interventions have thus been designed to modify, diminish or enhance autistic children's behaviours to address these key goals.

The most ubiquitous of these treatments are those that are based on a behavioural model, especially Applied Behavioural Analysis (ABA) (Vismara & Rogers, 2010). The ambition of ABA to render autistic children 'indistinguishable from their normal friends' (Lovaas, 1987, p. 8) at the very least obscures any distinctive autistic strengths. These treatments are controversial for a range of reasons (e.g., Dawson, 2004; Ne'eman, 2021; Wilkenfeld & McCarthy, 2020) but, for the purposes of this paper, it is helpful to concentrate on the treatment of one behaviour, stimming. Seen through the lens of the conventional medical model, there has been a tendency to perceive so-called repetitive motor stereotypies such as hand-flapping (American Psychiatric Association, 2013) as an individual problem, with no clear purpose or function, and which prevent the child from learning adaptive or other skills and interacting with their peers (Asperger, 1944; DiGennaro Reed, Hirst, & Hyman, 2012; Kanner, 1943; Lilley, 2018). Given these presumed negative outcomes – and that stimming behaviours like hand-flapping are considered conspicuous and stigmatising, particularly for children in mainstream classroom settings – it is often regarded as 'essential that researchers continue to assess and treat motor stereotypy either as the target behaviour of the intervention or as a collateral behaviour' (Tereshko, Ross, & Frazee, 2021, p. 3); to demonstrate 'calm' or 'quiet' hands.

These treatments persist despite a dearth of evidence that these stimming behaviours are harmful to autistic people or their peers. In fact, it now seems likely that they often serve a regulatory, soothing function for autistic people (e.g. Joyce, Honey, Leekam, Barrett, & Rodgers, 2017; Kapp et al., 2019). Moreover, the interventions designed to quash these behaviours might well do more harm than good (Bascom, 2012; Dawson, 2004; Jaswal & Akhtar, 2018; Lilley, 2018). It is surely at least plausible to suggest that attention would be better directed towards *social* interventions that aim to shift the negative perceptions of stimming among nonautistic people, rather than to efforts to prevent autistic people from stimming in the first place.

Excluded voices, narrow perspectives

One further critique of the conventional medical paradigm is its tendency to detract attention from autistic people's own understanding of autism and of their own lives. As autistic self-advocate Donna Williams put it, 'right from the start, from the time someone came up with the word "autism", the condition has been judged from the outside, by its

特别是针对年轻孤独谱系群体的研究发现，他们常常以一种消极的方式将自己视为与神经典型群体“不同”（Cribb、Kenny 与 Pellicano，2019；Humphrey 与 Lewis，2008；Shattuck 等人，2014；Williams、Gleeson 与 Jones，2019）。他们将自己标签化为“怪胎”、“心智残障”和“大脑有问题”（Humphrey 与 Lewis, 2008），并且常常只想“尽可能正常”，主动隐藏自己的孤独谱系特质（Cribb 等人，2019）。这些负面的自我评价并非意料之外，因为年轻孤独谱系群体更可能经历社会排斥和欺凌（Humphrey 与 Hebron，2014；Maiano、Normand、Salvas、Moullec 与 Aime，2016；Schroeder、Cappadocia、Bebko、Pepler 与 Weiss，2014）。但这可能对他们的心理健康造成损害（Cage、Di Monaco 与 Newell，2018；Cooper、Smith 与 Russell，2017；Hedley 与 Young，2006），其中残障/孤独谱系个体可能开始相信自己价值较低并据此行事（Becker，1963；Milton，2012）。通过掩饰或伪装主动隐藏自己的孤独谱系差异（相关讨论参见 Pearson 与 Rose，2021）也可能带来严重的个人代价，包括压力、焦虑和负面的自我认知（Hull 等人，2017），以及成年孤独谱系个体的倦怠（Higgins、Arnold、Weise、Pellicano 与 Trollor，2021；Raymaker 等人，2020）。

医学模式本质上是个人主义的

基于缺陷的思维方式的盛行，进一步导致了注意力直接聚焦于个体本身，而忽视了可能实际上在塑造孤独谱系群体的生活方面扮演重要角色的社会和环境因素（Engel，1977）。在传统的医学观点中，孤独谱系及其相关的障碍被视为个体固有的特质。因此，生物医学研究倾向于解释孤独谱系人士的困难时，不是参照困难发生的背景——家庭、学校、工作或更广泛的社区——而是将其视为个体自身的特征。在孤独谱系科学内部，这种倾向因孤独谱系人士被描绘为独自存在于世界中的既定形象而进一步加强（Broderick & Ne'eman, 2008）。鉴于孤独谱系最初被视为表现为从世界中退缩（Kanner, 1943），早期的研究者更直觉性地认为，决定孤独谱系体验本质的特征在于个体自身。

从这种个体化的出发点来看，生活中的困难“过错”被认为在于个体自身，因此“纠正”这些感知到的困难的负担也落在了个体身上。孤独谱系群体被认为在某种程度上存在“缺陷”或“不足”，需要个体治疗来弥补这些缺点。在孤独谱系研究和实践中，显而易见的是，孤独谱系群体总体上被期望“克服”他们的障碍，以达到典型的能力水平。因此，治疗和干预措施的设计旨在修改、减少或增强孤独谱系儿童的行为，以实现这些关键目标。

其中最为普遍的治疗方法是基于行为模型的疗法，尤其是应用行为分析（ABA）（Vismara & Rogers, 2010）。ABA的目标是让孤独谱系儿童“与他们的正常朋友无异”（Lovaas, 1987, p. 8），这至少掩盖了任何独特的孤独谱系优势。这些治疗方法因多种原因存在争议（例如，Dawson, 2004; Ne’eman, 2021; Wilkenfeld与McCarthy, 2020），但就本文而言，专注于一种行为——自我刺激行为的治疗是有益的。从传统医学模式的视角来看，人们倾向于将所谓的重复性运动刻板行为，如拍手（美国精神医学学会, 2013），视为个体问题，认为其没有明确的目的或功能，并阻碍儿童学习适应性或其他技能以及与同伴互动（Asperger, 1944; DiGennaro Reed, Hirst, & Hyman, 2012; Kanner, 1943; Lilley, 2018）。鉴于这些假定的负面后果——以及像拍手这样的自我刺激行为被认为引人注目且带有污名，尤其对于主流课堂环境中的儿童——它常被视为“研究者必须继续评估和治疗运动刻板行为，无论是作为干预的目标行为还是附带行为”（Tereshko、Ross与Frazee, 2021, p. 3）；以展示“平静”或“安静”的手部状态。

尽管缺乏证据表明这些自我刺激行为对孤独谱系群体或其同伴有害，但这些治疗手段依然持续存在。事实上，如今看来，这些行为很可能常常对孤独谱系群体起到调节、安抚的作用（例如 Joyce、Honey、Leekam、Barrett与Rodgers, 2017; Kapp等, 2019）。此外，旨在压制这些行为的干预措施很可能弊大于利（Bascom, 2012; Dawson, 2004; Jaswal与Akhtar, 2018; Lilley, 2018）。至少可以肯定地提出，注意力应更好地转向旨在改变非孤独谱系人群对自我刺激行为的负面看法，而非首先致力于阻止孤独谱系群体进行自我刺激行为的社会干预措施。

被排斥的声音，狭隘的视角

对传统医学范式的另一项批评在于，它倾向于忽视孤独谱系群体自身对孤独谱系及其生活的理解。正如孤独谱系自我倡导者 Donna Williams 所言：‘从一开始，从有人提出“孤独谱系”这个词起，这种状况就一直是从外部、根据其

appearances, and not from the inside according to how it is experienced' (Williams, 1996, p. 14). Along these lines, some researchers have even suggested that a lack of 'theory of mind' in autistic children, young people and adults (Baron-Cohen, Leslie, & Frith, 1985) impairs their ability to reflect on their own mental states (Lombardo, Chakrabarti, Bullmore, & Baron-Cohen, 2011), thus questioning the veracity of autistic people's accounts of their own experiences (Frith & Happé, 1999).

Consequently, researchers have often avoided attending to first-person testimony, preferring to privilege reports from parents, teachers or other informants, or laboratory-based observation over considering the perspectives of the person themselves (Jaswal & Akhtar, 2018; Mazefsky, Kao, & Oswald, 2011; McGeer, 2004; Milton, 2012). The claim that autistic people cannot understand their own minds can also have damaging effects, including to autistic people's autonomy, self-determination and perceived credibility (Gernsbacher & Yergeau, 2019). As Jim Sinclair (1993, p. 298) wrote:

> The results of these assumptions are often subtle, but they're pervasive and pernicious: I am not taken seriously. My credibility is suspect. My understanding of myself is not considered to be valid, and my perceptions of events are not considered to be based in reality. My rationality is questioned because, regardless of intellect, I still appear odd. My ability to make reasonable decisions, based on my own carefully reasoned priorities, is doubted because I don't make the same decisions that people with different priorities would make.

As well as shaping research findings, this lack of attention to autistic people's perspectives also has the consequence of ensuring that autistic people themselves have almost no say as to *what* gets researched in autism science, *why* or *how*. Over the last two decades, international investment in autism science has grown extensively (Office of Autism Research Coordination, 2017; Pellicano, Dinsmore, & Charman, 2013). In 2016 alone, US$400 million was spent on autism research in the United States, United Kingdom, Australia and Canada – although the United States contributed the vast majority (92%) of that investment (Office of Autism Research Coordination, 2019). Similarly, the number of papers published on autism has increased 10-fold (Dawson, 2013; Office of Autism Research Coordination, 2012), far surpassing publications on related topics in child psychology and psychiatry. According to PubMed, there were 6,539 journal articles published on autism in 2020 alone.

Autistic people themselves have had almost no say in shaping that research agenda. The conventional medical approach to autism assumes that priorities for autism research centre on questions of identifying, treating and potentially preventing autism; and that answers to these fundamental questions will 'only be found when there is a better understanding of the neurobiological basis of autism' (Insel & Daniels, 2011, p. 1361). As noted by Bertilsdotter Rosqvist and Jackson-Perry (2020), 'in a condition as highly medicalised as autism, whose parameters are laid out in medical practitioners' diagnostic manuals, it follows that the testimony which may be assumed to be the most accurate, which holds the highest level of credibility when discussing autism… issues from medical science' (p. 3).

It is not, surprising, then, that the vast preponderance of autism research worldwide focuses on the underlying genetic causes and biology of autism (den Houting & Pellicano, 2019; Office of Autism Research Coordination, 2019; Pellicano, Dinsmore, & Charman, 2014b; Singh, Illes, Lazzeroni, & Hallmayer, 2009). This focus is in sharp contrast to the stated research priorities of community members – autistic people, their family members, educators, clinicians and other professionals – who have consistently called for research on areas that are of more immediate, practical concern or for basic science research that may be more straightforwardly translated into applications (Frazier et al., 2018; Jose et al., 2020; Pellicano et al., 2014b; Robertson, 2010; Roche, Adams, & Clark, 2021; Warner, Parr, & Cusack, 2019). In the words of one parent, 'when it comes down to it, it's not real life … [research] is always missing the next step' (Pellicano et al., 2014b, p. 5).

The relationship between autism scientists and community members has a long and fraught history (Pellicano & Stears, 2011; Silberman, 2015; Silverman, 2011). But the mismatch between what is currently being researched, what community members want from research and who gets to make these decisions may well be contributing to the growing distrust of mainstream autism science by the broader community (Bagatell, 2010; Dawson, 2004; Lory, 2019; Milton, 2012). Pellicano, Dinsmore, and Charman (2014a) have shown that autistic people and family members in the United Kingdom report overwhelmingly negative experiences of autism research. Family members described feeling disappointed and frustrated at being 'mined' for information and having little or no opportunity to learn about the resulting scientific discoveries and what they might mean for them. Autistic adults reported feeling objectified ('we are a bit like monkeys in a zoo') and their experiential expertise perceived as being disregarded by researchers ('whatever I say, is this really going to influence anyone?'; also see Milton & Bracher, 2013).

Ultimately, autism research has been characterised by a narrowness of perspective. By prioritising research on causation, we have failed to understand the nature of autistic people's life experiences. Imposing nonautistic, medically driven priorities has had the grave consequence of diverting resources away from existing autistic people and the

表象来判断，而非从内部依据其实际体验来判断’（Williams, 1996, p. 14）。沿着这一思路，一些研究者甚至提出，孤独谱系儿童、青少年和成人缺乏心理理论（Baron-Cohen, Leslie, & Frith, 1985），这削弱了他们反思自身心理状态的能力（Lombardo, Chakrabarti, Bullmore, & Baron-Cohen, 2011），从而质疑了孤独谱系群体对其自身体验描述的准确性（Frith & Happ´e, 1999）。

因此，研究者们常常回避关注第一人称证词，更倾向于优先采纳来自父母、教师或其他信息提供者的报告，或基于实验室的观察，而非考虑当事人自身的视角（Jaswal与Akhtar, 2018; Mazefsky, Kao, & Oswald, 2011; McGeer, 2004; Milton, 2012）。声称孤独谱系群体无法理解自己的心智，也可能产生破坏性影响，包括损害孤独谱系群体的自主性、自决以及被感知的可信度（Gernsbacher & Yergeau, 2019）。正如Jim Sinclair（1993, p. 298）写道：

> 这些假设所导致的结果往往微妙，但却无处不在且有害：我不被认真对待。我的可信度受到怀疑。我对自身的理解不被认为是有效的，我对事件的感知不被认为是基于现实的。我的理性受到质疑，因为无论智力如何，我仍然显得古怪。我基于自身仔细推理出的优先级做出合理决策的能力受到怀疑，因为我没有做出那些拥有不同优先级的人会做出的决定。

除了影响研究成果外，这种对孤独谱系群体视角的忽视也导致孤独谱系群体自身几乎无法决定孤独谱系科学中研究什么、为何研究以及如何研究。在过去二十年里，国际上对孤独谱系科学的投入大幅增长（孤独症研究协调办公室，2017；Pellicano, Dinsmore, & Charman, 2013）。仅2016年，美国、英国、澳大利亚和加拿大在孤独谱系研究上就花费了4亿美元——尽管美国贡献了其中绝大部分（92%）的投资（孤独症研究协调办公室，2019）。同样，关于孤独谱系的论文发表数量增长了10倍（Dawson, 2013；孤独症研究协调办公室，2012），远超儿童心理学和精神病学相关主题的出版物。根据PubMed的数据，仅2020年就有6,539篇关于孤独谱系的期刊文章发表。

孤独谱系人士自身在塑造该研究议程方面几乎没有发言权。针对孤独谱系的传统医学方法假定，孤独谱系研究的优先事项集中在识别、治疗和潜在预防孤独谱系的问题上；并且认为这些基本问题的答案‘只有在更好地理解孤独谱系的神经生物学基础时才能找到’（Insel & Daniels, 2011, p. 1361）。正如Bertilsdotter Rosqvist and Jackson-Perry（2020）所指出的，‘在像孤独谱系这样高度医学化的状况中，其参数由医疗从业者的诊断手册所规定，随之而来的是，在讨论孤独谱系…问题时，被认为最准确、可信度最高的证言来自医学科学’（p. 3）。

因此，全球范围内绝大多数孤独谱系研究都聚焦于孤独谱系的潜在遗传成因与生物学基础（den Houting与Pellicano, 2019；孤独症研究协调办公室, 2019；Pellicano, Dinsmore, & Charman, 2014b；Singh, Illes, Lazzeroni, & Hallmayer, 2009），这并不令人意外。这一焦点与社区成员——孤独谱系群体、他们的家庭成员、教育工作者、临床医生及其他专业人士——所明确提出的研究优先事项形成了鲜明对比。他们持续呼吁研究应关注更具直接实践意义的领域，或开展那些可能更易于转化为应用的基础科学研究（Frazier et al., 2018；Jose et al., 2020；Pellicano et al., 2014b；Robertson, 2010；Roche, Adams, & Clark, 2021；Warner, Parr, & Cusack, 2019）。正如一位家长所言：“归根结底，这不是真实的生活 … [研究] 总是缺失下一步”（Pellicano et al., 2014b, p. 5）。

孤独谱系科学家与社区成员之间的关系有着漫长且充满矛盾的历史（Pellicano & Stears, 2011; Silberman, 2015; Silver- man, 2011）。然而，当前的研究内容、社区成员对研究的期望以及由谁来做出这些决策之间的不匹配，很可能加剧了广大社区对主流孤独谱系科学日益增长的不信任感（Bagatell, 2010; Dawson, 2004; Lory, 2019; Milton, 2012）。Pellicano、Dinsmore和Charman（2014a）的研究表明，英国的孤独谱系群体及其家庭成员报告了对孤独谱系研究压倒性的负面体验。家庭成员描述了他们感到失望和沮丧，因为他们感觉自己被‘挖掘’信息，却几乎没有机会了解由此产生的科学发现以及这些发现对他们可能意味着什么。成年孤独谱系个体报告了他们感觉自己被物化（‘我们有点像动物园里的猴子’），并且他们的经验性专业知识被认为被研究人员所忽视（‘无论我说什么，这真的会影响任何人吗？’；另见Milton & Bracher, 2013）。

归根结底，孤独谱系研究一直因视角狭隘而备受诟病。由于优先研究成因，我们未能理解孤独谱系群体生活体验的本质。强加非孤独谱系、医学驱动的优先事项，已导致资源从现有的孤独谱系群体及其

areas that matter most in their lives (Pellicano & Stears, 2011; Robertson, 2010).

Neurodiversity: is it time for a paradigm shift?

These three difficulties with the conventional medical approach to autism – an overfocus on deficits, an overwhelming emphasis on the individual as opposed to their social context and a narrowness of perspective – have provoked an increasingly strong reaction in recent years. In the early 1990s, autistic activists began to advocate for a shift in attitudes towards autistic people. They drew encouragement from the broader disability rights movement that itself gained force during the late 1970s and early 1980s, and achieved landmark progress during the early 1990s with the introduction of the Americans with Disabilities Act in 1990, the Australian Disability Discrimination Act in 1992, and the British Disability Discrimination Act in 1995 (Oliver, 1996; Swain, French, & Cameron, 2003). Perhaps most notable in this early autistic advocacy is the work of autistic advocate Jim Sinclair, whose foundational essay 'Don't mourn for us' remains poignantly relevant even today (Pripas-Kapit, 2020; Sinclair, 1993).

In 1998, autistic sociologist Judy Singer and journalist Harvey Blume coined the word 'neurodiversity' (Blume, 1998; Singer, 1998), supplying a collective framework for autistic and neurodivergent advocacy. The term 'neurodiversity' (see Box 1 for terminology) and the associated paradigm were embraced by autistic advocates, and over time became ubiquitous within – and synonymous with – the broader neurodivergent community. Throughout the 2000s, autistic-led advocacy organisations (e.g. the Autistic Self-Advocacy Network) were founded, providing for the first time an authoritative voice for the autistic community in neurodiversity advocacy efforts.

Autistic scholars and allies have worked to promote awareness of the neurodiversity paradigm within academia (den Houting, 2019; Nicolaidis, 2012; Robertson, 2010). A cursory search of the PubMed database using the keyword 'neurodiversity' indicates some success in this regard. During 2010, just one relevant manuscript was published. A decade later during 2020, 33 manuscripts were published. Most recently, autistic scholars have sought to progress the scholarship of neurodiversity in its own right, through the emerging disciplines of critical autism studies and neurodiversity studies (Bertilsdotter Rosqvist, Chown, & Stenning, 2020; Kapp, 2020; Walker & Raymaker, 2020). Framed within the neurodiversity paradigm, each critique explored above is seen as a significant failure requiring urgent attention to put right. Below, we trace the ways in which those who advocate neurodiversity seek to do so.

Box 1 Terminology

Neurodiversity: The range of natural diversity that exists in human neurodevelopment.
Neurotypical: A person or people whose neurodevelopment falls within the range usually considered to constitute 'typical' development.
Neurodivergent: A person or people whose neurodevelopment falls outside of (or 'diverges' from) the range usually considered to constitute 'typical' development (e.g. a group of autistic people is a group of 'neurodivergent' people).
Neurodiverse: A collective term for groups including mixed neurodevelopment (e.g. a group of autistic and nonautistic people is a 'neurodiverse' group).

Diversity, not deficits

In and of itself, *neurodiversity* refers to the broad diversity that exists in human neurobiology. There are countless ways in which the human brain and mind can develop, both structurally and functionally. Many of these fall within a range that can be considered as 'typical' neurodevelopment, while some fall outside of that range and can be considered to 'diverge' from the norm (Ecker, Bookheimer, & Murphy, 2015; Jumah, Ghannam, Jaber, Adeeb, & Tubbs, 2016). Neurodiversity encompasses this entire spectrum, including both typical and divergent neurodevelopment (Singer, 1998; Walker, 2014). The neurodiversity paradigm refers to a particular set of beliefs and attitudes regarding this diversity (Walker, 2012) and rejects the view that divergence from the norm is a flaw requiring correction.

This stance can be broken down into two key assumptions. The first of these is the assumption that typical neurodevelopment is neither superior nor inferior to divergent neurodevelopment. Neurotypicality does not represent 'correct' neurodevelopment any more than English represents the 'correct' language to speak, or that one element of an ecosystem can be seen as prior to any other. Indeed, diversity in neurodevelopment is itself valuable, in much the same way that linguistic diversity is valuable in contributing to a culturally rich landscape or biological diversity can strengthen the health of all of the component parts of a system (see Amundson, 2000, for discussion). The second key assumption within the neurodiversity paradigm is the belief that even if diversity were not to serve this collective purpose, *all* people deserve to be treated with dignity and respect, independently of how they diverge from a putative norm, and should be valued for who they are and as they are.

This second assumption implies that, in contrast to the common misconception that the neurodiversity

生活中最重要的领域被转移（Pellicano & Stears, 2011; Robertson, 2010）。

神经多样性：是否到了范式转变的时刻？传统医学方法对待孤独谱系的这三个难题——过度聚焦于缺陷、压倒性地强调个体而非其社会背景以及视角的狭隘性——近年来引发了日益强烈的反应。早在1990年代初，孤独谱系活动家就开始倡导对孤独谱系群体的态度转变。他们从更广泛的残障权利运动中汲取鼓舞，该运动本身在1970年代末和1980年代初获得力量，并在1990年代初取得了里程碑式的进展，包括1990年《美国残障人士法案》、1992年《澳大利亚残障歧视法案》和1995年《英国残障歧视法案》的引入（Oliver, 1996; Swain, French, & Cameron, 2003）。或许在这早期的孤独谱系倡导中，最值得注意的是孤独谱系倡导者Jim Sinclair的工作，其奠基性文章《不要为我们哀悼》至今仍切中要害，令人动容（Pripas-Kapit, 2020; Sinclair, 1993）。

1998年，孤独谱系社会学家Judy Singer和记者Harvey Blume创造了‘神经多样性’一词（Blume, 1998; Singer, 1998），为孤独谱系及神经殊异倡导提供了一个集体框架。‘神经多样性’这一术语（术语定义见方框1）及相关范式被孤独谱系倡导者所接纳，并随着时间的推移，在更广泛的神经殊异社群中变得无处不在，甚至成为其同义词。在整个2000年代，孤独谱系主导倡导组织（例如孤独谱系自我倡导网络）相继成立，首次在神经多样性倡导运动中为孤独谱系社群提供了权威的声音。

孤独谱系学者及其盟友致力于在学术界推广神经多样性范式的认知（den Houting, 2019; Nicolaidis, 2012; Robertson, 2010）。在PubMed数据库中，以“神经多样性”为关键词进行粗略检索，表明在这方面已取得一定成效。2010年，仅有一篇相关论文发表。十年后的2020年，则有33篇论文发表。最近，孤独谱系学者正试图通过新兴的批判孤独谱系研究和神经多样性研究学科，推动神经多样性学术本身的发展（Bertilsdotter Rosqvist, Chown, & Stenning, 2020; Kapp, 2020; Walker & Raymaker, 2020）。在神经多样性范式的框架下，上文探讨的每一项批评都被视为亟待纠正的重大缺陷。下文我们将追溯倡导神经多样性者为实现此目标所采取的方式。

框1 术语

神经多样性：人类神经发育中存在的自然多样性范围。

神经典型：其神经发育处于通常被认为是构成“典型”发育范围内的人或人群。

神经殊异：指神经发育超出（或‘偏离’）通常被视为‘典型’发育范围的人或人群（例如，一群孤独谱系群体即是一群‘神经殊异’人群）。神经多元：指包含混合神经发育群体的集合术语（例如，一群孤独谱系和非孤独谱系人群即是一个‘神经多元’群体）。

多样性，而非缺陷

就其本身而言，神经多样性指的是人类神经生物学中存在的广泛多样性。人脑和心智在结构和功能上的发育方式不计其数。其中许多方式属于可被视为‘典型’神经发育的范围，而有些则超出该范围，可被视为与常态‘殊异’（Ecker, Bookheimer, & Murphy, 2015; Jumah, Ghannam, Jaber, Adeeb, & Tubbs, 2016）。神经多样性涵盖了这整个谱系，包括典型和殊异的神经发育（Singer, 1998; Walker, 2014）。神经多样性范式指的是一套关于这种多样性的特定信念和态度（Walker, 2012），并拒绝将偏离常态视为需要纠正的缺陷的观点。

这一立场可分解为两个关键假设。第一个假设是，典型的神经发育既不优于也不劣于非典型的神经发育。神经典型性并不代表‘正确’的神经发育，正如英语不代表‘正确’的说话语言，或生态系统中的一个元素不能被视作优先于其他任何元素。事实上，神经发育的多样性本身是有价值的，其价值在很大程度上类似于语言多样性有助于形成文化丰富的景观，或生物多样性能够增强系统所有组成部分的健康（参见 Amundson, 2000 的讨论）。神经多样性范式中的第二个关键假设是，即使多样性不服务于这一集体目的，所有人也应享有尊严和尊重，无论他们如何偏离一个假定的规范，并且应因其本身的样子而被重视。

第二个假设意味着，与常见的误解——即神经多样性

movement is only of interest to autistic people with less complex support needs (commonly termed 'high functioning'; Fenton & Krahn, 2007; Jaarsma & Welin, 2012), the neurodiversity paradigm is explicitly inclusive of *all* autistic and neurodivergent people, including those with the highest and most complex support needs (Grinker, 2020; den Houting, 2019; Ne'eman, 2021). In practice, neurodiversity advocates have long recognised that it can be a challenge to ensure that all autistic people's interests are taken into account, not just those who are more able to influence decision-making or to make their voices heard through research, activism or protest (Ballou, 2018). But, fundamentally, those who support neurodiversity accept that a person's value must not be contingent on their ability to 'contribute' to society, economically or otherwise (Kittay, 2017). As such, the neurodiversity paradigm also broadens the standard understanding of what constitutes a 'meaningful' life. It insists that an autistic person's life may be rich and fulfilling despite bearing little resemblance to the conventional ideal. In conventional research, well-being and quality of life are often presented as objective constructs comprising outcomes including employment, independence, and relationships. Neurodiversity, on the other hand, encourages us to be open to a wider range of potential conceptualisations of a good life. Measured according to neuronormative standards, that is, many autistic people are assumed to have poor quality of life, but this view can change if autism-specific values, goals and needs are taken into account (Lam, Sabnis, Migueliz Valcarlos, & Wolgemuth, 2021). Indeed, when autistic people have been involved in determining outcomes that are relevant to their quality of life, they have identified unique factors including other people's knowledge and acceptance of autism; sensory processing differences; supporting other people and positive autistic identity (McConachie et al., 2020). This contrasts sharply with deficit-based framings of autism as an unwelcome encumbrance (e.g. autism as a 'shell' or 'prison'; Broderick & Ne'eman, 2008) from which an otherwise 'normal' person can be freed through intervention.

The neurodiversity paradigm thus promotes autism acceptance, urging autistic people and others to embrace autism as an inherent and integral part of an autistic person's identity and experience of the world. The potential value of this acceptance mindset to autism is further supported by recent research indicating that greater autism acceptance is associated with better mental health, in both autistic adults (Cage et al., 2018) and mothers of autistic children (Da Paz, Siegel, Coccia, & Epel, 2018). Similarly, and consistent with evidence regarding identity development in the broader disabled population and the general population, a positive sense of autistic identity is also associated with better mental health (Cooper et al., 2017).

Neurodiversity highlights the need for social responses

None of this is to suggest, of course, that autistic people do not require supports to enable them to achieve their goals. The neurodiversity paradigm does, however, lead us to understand the nature of the obstacles that many autistic people face in a fundamentally different way to the conventional medical paradigm. Neurodiversity is closely aligned with the social model of disability. Seen this way, any 'disability' is not best understood as the result of an individual's unique characteristics, but rather as a result of an environment that does not effectively accommodate those characteristics (Oliver, 1996; Union of the Physically Impaired Against Segregation, 1976). The physical and social environments within which we all live are generally designed to meet the needs of those who fall within the typical range of neurodevelopment; these same environments are often suboptimal, and even hostile, for neurodivergent people and need to be adjusted if such people are to lead good lives.

Traditional, mainstream school settings are an example of one such environment: they are often physically large, noisy and chaotic, require frequent transitions within and between classes throughout the school day, and entail a host of implicit social rules and expectations. Physical environments, like schools, that are designed with neurotypical needs in mind can leave autistic people in sensory discomfort and distress, and social environments governed by neurotypical rules are often similarly inaccessible. These inaccessible contexts can render autistic people less 'able' than their neurotypical peers – that is, disabled – in myriad ways.

Such examples are only small instances of the ways in which the institutions that govern society at a structural level are typically controlled by and designed for neurotypical people, often with devastating effects for autistic people. Autistic people are disproportionately formally excluded (expelled) from school (e.g. Ambitious about Autism, 2014; Brede, Remington, Kenny, Warren, & Pellicano, 2017), experience frequent bullying and other forms of victimisation (Brown-Lavoie, Viecili, & Weiss, 2014; Maiano et al., 2016), are either unemployed or underemployed at greater rates than other disabled people (Chen, Leader, Sung, & Leahy, 2014; Scott et al., 2019), are at greater risk of physical health conditions (Croen et al., 2015) and have increased vulnerability to mental ill-health (Lai et al., 2019), including high rates of suicide (Cassidy et al., 2014; Kirby et al., 2019). They are also more likely to experience premature death (Hirvikoski et al., 2016). A growing body of research describes the substantial barriers that autistic adults face in accessing physical healthcare, which include such diverse factors as inaccessible sensory environments; providers'

该运动仅对支持需求较简单的孤独谱系群体（通常称为‘高功能’；Fenton & Krahn, 2007；Jaarsma & Welin, 2012）感兴趣，而神经多样性范式明确涵盖所有孤独谱系和神经殊异人群，包括那些支持需求最高且最复杂的人群（Grinker, 2020；den Houting, 2019；Ne'eman, 2021）。在实践中，神经多样性倡导者早已认识到，确保所有孤独谱系群体的利益都被纳入考量是一项挑战，不仅仅是那些更有能力影响决策或通过研究、行动主义或抗议发声的人群（Ballou, 2018）。但根本上，支持神经多样性的人士认同，一个人的价值不应取决于其能否在经济或其他方面为社会‘做出贡献’（Kittay, 2017）。因此，神经多样性范式也拓宽了对何为‘有意义’生活的标准理解。它坚持认为，孤独谱系个体的生活即使与传统理想相差甚远，也可能丰富而充实。在传统研究中，福祉和生活质量常被呈现为客观建构，包含就业、独立性和人际关系等结果。另一方面，神经多样性鼓励我们对美好生活的潜在概念化持更开放的态度。按照神经典型性标准来衡量，许多孤独谱系群体被认为生活质量较差，但如果考虑到孤独谱系特有的价值观、目标和需求，这种观点可以改变（Lam, Sabnis, Migueliz Valcarlos, & Wolgemuth, 2021）。事实上，当孤独谱系群体参与确定与其生活质量相关的结果时，他们识别出了独特的因素，包括他人对孤独谱系的了解和接纳；感官处理差异；支持他人以及积极的孤独谱系身份认同（McConachie et al., 2020）。这与将孤独谱系视为不受欢迎负担的缺陷型框架（例如，将孤独谱系视为‘外壳’或‘监狱’；Broderick & Ne'eman, 2008）形成鲜明对比，后者认为一个原本‘正常’的人可以通过干预从中解脱出来。

因此，神经多样性范式倡导孤独谱系接纳，敦促孤独谱系群体及其他人士将孤独谱系视为孤独谱系个体身份与世界体验中固有且不可或缺的一部分。这种接纳心态对孤独谱系的潜在价值，进一步得到了近期研究的支持，表明更高的孤独谱系接纳与更好的心理健康相关，无论是在成年孤独谱系个体（Cage et al., 2018）还是孤独谱系儿童的母亲中（Da Paz, Siegel, Coccia, & Epel, 2018）。同样地，并且与关于更广泛残障人群及普通人群身份发展的证据相一致，积极的孤独谱系身份认同也与更好的心理健康相关（Cooper et al., 2017）。

神经多样性凸显了社会回应的必要性

当然，这并非暗示孤独谱系群体不需要支持来实现其目标。然而，神经多样性范式确实引导我们以一种与传统医学范式根本不同的方式，去理解许多孤独谱系群体所面临的障碍的本质。神经多样性范式与残障社会模式紧密契合。以此视角来看，任何‘残障’最好不被理解为个体独特特征的结果，而应理解为环境未能有效适应这些特征的结果（Oliver, 1996；Union of the Physically Impaired Against Segregation, 1976）。我们所有人生活的物理和社会环境，通常是为满足那些处于典型神经发育范围内人群的需求而设计的；这些同样的环境对于神经殊异群体而言往往并非最优，甚至充满敌意，需要进行调整，以使此类群体能够过上良好的生活。

传统的主流学校环境便是此类环境的一个例子：它们通常在物理空间上规模庞大、嘈杂混乱，要求学生在整个上学日频繁地在课堂内外进行转换，并包含大量隐含的社会规则和期望。像学校这样为满足神经典型需求而设计的物理环境，可能会让孤独谱系群体感到感官不适和痛苦；而由神经典型规则主导的社会环境，也往往同样难以融入。这些难以融入的环境，会以无数种方式使孤独谱系群体显得比其神经典型同伴更“无能”——也就是说，成为残障。

这些例子只是冰山一角，它们揭示了在结构层面治理社会的机构通常由神经典型群体控制并为其设计，而这往往对孤独谱系群体造成毁灭性影响。孤独谱系群体在学校中被不成比例地正式排除（开除）（例如 Ambitious about Autism, 2014; Brede, Remington, Kenny, Warren, & Pellicano, 2017），频繁遭受欺凌和其他形式的受害（Brown-Lavoie, Viecili, & Weiss, 2014; Maiano et al., 2016），失业或就业不足的比例高于其他残障群体（Chen, Leader, Sung, & Leahy, 2014; Scott et al., 2019），面临更高的身体健康问题风险（Croen et al., 2015），并且对心理健康问题的脆弱性增加（Lai et al., 2019），包括高自杀率（Cassidy et al., 2014; Kirby et al., 2019）。他们也更可能经历过早死亡（Hirvikoski et al., 2016）。越来越多的研究描述了成年孤独谱系个体在获取物理医疗服务时面临的巨大障碍，这些障碍包括诸如难以接近的感官环境；服务提供者的

knowledge about and attitudes towards autism; and the complexity of healthcare systems (Mason et al., 2019; Nicolaidis et al., 2015). Similar barriers exist across a range of domains, including access to mental healthcare (Adams & Young, 2020), employment (Harmuth et al., 2018) and leisure activities (Askari et al., 2014). According to the neurodiversity paradigm, responding to these facts does not require us to 'change autistic people' but rather to challenge the societal factors that influence these outcomes (Howlin, 2000; Howlin & Magiati, 2017; see Mandy et al., 2016, for an example).

The barriers to autistic people's flourishing include both those that exist in the physical environment *and*, perhaps more importantly, social and attitudinal barriers. The discrimination and stigma described above constitutes a serious obstacle for autistic people, which can only be addressed at the societal and systemic levels by promoting autistic inclusion and equity.

Autistic people should lead the conversation

This point returns us to the issue raised above concerning the exclusion of autistic people themselves from the decision-making that directs the agenda for much autism research. It is possible that this exclusion in part results from the way in which we conventionally conceive the fundamental relationship between autistic and nonautistic people. Milton's (2012) influential Double Empathy Problem proposes a misalignment between the minds of autistic and nonautistic people, highlighting a lack of reciprocity in cross-neurotype interactions as the source of social communication difficulties between autistic and nonautistic people (Davis & Crompton, 2021; Milton, Heasman, & Sheppard, 2018; Mitchell, Sheppard, & Cassidy, 2021). What such research shows is that *nonautistic* people have difficulties understanding the minds and behaviours of autistic people – and, worryingly, are prone to misperceiving them. Nonautistic people, for example, struggle to understand autistic people's facial expressions (e.g. Brewer et al., 2016) and have difficulties interpreting autistic people's behaviour (Sheppard, Pillai, Wong, Ropar, & Mitchell, 2016) and mental states (Edey et al., 2016). Nonautistic people also report an unwillingness to interact with autistic people based on first impression judgements (Morrison, DeBrabander, Faso, & Sasson, 2019; Sasson et al., 2017) and brief interactions (Morrison et al., 2020).

Consistent with this hypothesis, autistic people have been found to communicate more effectively (Crompton, Ropar, Evans-Williams, Flynn, & Fletcher-Watson, 2020) and develop stronger rapport (Crompton et al., 2020) with other autistic people than with nonautistic people. Emerging evidence also suggests that autistic social interaction may incorporate distinctive features that promote social understanding between autistic people (termed 'neurodivergent intersubjectivity'; Heasman & Gillespie, 2019).

The neurodiversity paradigm responds to the apparent lack of alignment between the minds of autistic and nonautistic people by emphasising the importance of self-determination and autonomy: autistic people must be involved in all decision-making that stands to affect autistic people, from the highest levels of policy development to individual support planning; that is, they 'deserve a full seat at the main table' (Gernsbacher, 2007, p. 13). With this in mind, neurodiversity advocates assert that autism research and practice must be brought into line with the needs and priorities of the autistic and autism communities, taking a far broader perspective of what matters in autism research as a result (Milton, 2014b; Raymaker, 2020; Robertson, 2010). As discussed above, however, autism science has not, to date, been reflective of community priorities for research.

One strategy for producing autism research that more closely aligns with community priorities is to involve autistic people and their allies directly in the research process, through participatory, codesigned and coproduced research. Community engagement in research has a long history outside of autism, most notably with regard to HIV (Epstein, 1996; Moore, 2008) and Aboriginal and First Nations communities (National Health & Medical Research Council, 1991). Processes that draw on the 'practical wisdom' of nonscientists have the potential to make scientific discoveries more thoroughly relevant to communities, more directly tailored to the realities of their everyday lives and more consistent with their values (Chalmers, 2004; Hickey, 2018; Lloyd & White, 2011; Pellicano, 2020).

In the past decade, autism researchers worldwide have begun to engage with the autistic community, consulting with autistic people at all stages of the research process, exploring ways to work with autistic people as partners, and addressing research topics prioritised by the community (see Fletcher-Watson et al., 2019; Nicolaidis et al., 2011; Pellicano et al., 2014a, 2014b; Pellicano & Stears, 2011). Such involvement requires substantial commitment by individual researchers to listen to, and learn from, the experiential expertise of a diverse range of autistic people (Collins & Evans, 2002; Milton, 2014a), and to be willing to share power and control in the research process with nonscientists.

Commitments to autistic involvement in research cannot be left entirely up to individual researchers. They must also extend to institutions, such as funders and universities, to instigate structural and strategic change. Over the last few years, funding agencies in the United States (National Institutes of Health's Interagency Autism Coordinating Committee), the United Kingdom (Autistica) and Australia (Cooperative Research Centre for Living with Autism; hereafter 'Autism CRC') have

对孤独谱系的了解和态度；以及医疗系统的复杂性（Mason et al., 2019; Nicolaidis et al., 2015）。类似的障碍存在于一系列领域，包括获取心理健康服务（Adams & Young, 2020）、就业（Harmuth et al., 2018）和休闲活动（Askari et al., 2014）。根据神经多样性范式，应对这些事实并不需要我们‘改变孤独谱系群体’，而是挑战影响这些结果的社会因素（Howlin, 2000; Howlin & Magiati, 2017；参见 Mandy et al., 2016 的例子）。

阻碍孤独谱系群体蓬勃发展的因素既包括物理环境中存在的障碍，或许更重要的是社会与态度层面的障碍。上文所述的歧视与污名构成了孤独谱系群体的严重阻碍，唯有在社会与系统层面推动孤独谱系群体包容与公平才能解决此问题。

孤独谱系群体应主导对话

这一点让我们回到上文提及的问题：孤独谱系群体自身被排除在决定许多孤独谱系研究议程的决策过程中。这种排斥可能部分源于我们传统上对孤独谱系与非孤独谱系人群之间基本关系的理解。Milton（2012）提出的具有影响力的双向同理心问题指出孤独谱系与非孤独谱系人群心智之间的错位，强调跨神经类型互动中缺乏互惠性是两者社交沟通困难的根源（Davis与Crompton, 2021；Milton, Heasman, & Sheppard, 2018；Mitchell, Sheppard, & Cassidy, 2021）。此类研究表明，非孤独谱系人群难以理解孤独谱系群体的心智与行为——更令人担忧的是，他们倾向于误解孤独谱系群体。例如，非孤独谱系人群难以理解孤独谱系群体的面部表情（如 Brewer et al., 2016），在解读孤独谱系群体的行为（Sheppard, Pillai, Wong, Ropar, & Mitchell, 2016）与心理状态（Edey et al., 2016）方面也存在困难。非孤独谱系人群还表示，基于第一印象判断（Morrison, DeBrabander, Faso, & Sasson, 2019；Sasson et al., 2017）和短暂互动（Morrison et al., 2020），他们不愿与孤独谱系群体互动。

与此假设一致，研究发现孤独谱系群体与其他孤独谱系群体沟通时比与非孤独谱系群体沟通更有效（Crompton, Ropar, Evans-Williams, Flynn, & Fletcher-Watson, 2020），且能建立更强的融洽关系（Crompton et al., 2020）。新出现的证据也表明，孤独谱系群体社交互动可能包含促进孤独谱系群体间社会理解的独特特征（被称为‘神经殊异主体间性’；Heasman & Gillespie, 2019）。

神经多样性范式通过强调自决与自主性的重要性，来回应孤独谱系与非孤独谱系群体之间思维明显不一致的问题：孤独谱系群体必须参与所有可能影响孤独谱系群体的决策制定，从最高层面的政策制定到个人支持计划；也就是说，他们‘理应享有主桌的完整席位’（Gernsbacher, 2007, p. 13）。基于此，神经多样性倡导者主张，孤独谱系研究与实践必须与孤独谱系群体及孤独谱系社群的需求和优先事项保持一致，从而对孤独谱系研究中何为重要之事采取更广阔的视角（Milton, 2014b; Raymaker, 2020; Robertson, 2010）。然而，如上所述，迄今为止，孤独谱系科学并未反映出社群对研究的优先关切。

一种能产出更贴近社群优先事项的孤独谱系研究的策略是，通过参与式、协同设计与共同产出研究，让孤独谱系群体及其盟友直接参与研究过程。社群参与研究在孤独谱系领域之外有着悠久历史，最显著的是在HIV（Epstein, 1996; Moore, 2008）以及原住民与第一民族社群（国家健康与医学研究委员会, 1991）方面。借鉴非科学家‘实践智慧’的过程，有可能使科学发现更彻底地贴合社群需求，更直接地适应他们日常生活的现实，并更符合他们的价值观（Chalmers, 2004; Hickey, 2018; Lloyd & White, 2011; Pellicano, 2020 ）。

在过去十年中，全球的孤独谱系研究者开始与孤独谱系社群接触，在研究过程的各个阶段咨询孤独谱系群体，探索与孤独谱系群体作为合作伙伴共同工作的方式，并致力于解决由社群优先确定的研究课题（参见 Fletcher-Watson et al., 2019; Nicolaidis et al., 2011; Pellicano et al., 2014a, 2014b; Pellicano & Stears, 2011）。这种参与要求个体研究者做出实质性承诺，倾听并学习来自多样化孤独谱系群体的经验性专业知识（Collins & Evans, 2002; Milton, 2014a），并愿意在研究过程中与非科学家分享权力和控制权。

确保孤独谱系群体参与研究的承诺不能完全交由研究人员个人决定。这些承诺也必须延伸至机构层面，例如资助方和大学，以推动结构性及战略性的变革。过去几年中，美国（国立卫生研究院的跨部门孤独谱系协调委员会）、英国（Autistica）和澳大利亚（自闭症生活合作研究中心；以下简称‘自闭症CRC’）的资助机构已

responded by attempting to direct research efforts towards issues prioritised by community members, including identifying effective supports, interventions and services for autistic people and their families.

The Autism CRC's attempts in Australia have been particularly successful in this regard. They adopted a whole-of-life approach to autism research, investing in projects across early years, school years and adulthood, as well as a participatory approach, placing the autistic community at the centre of its research efforts. This strategic oversight paid off. Analysis of the funding expenditure of Australian autism research during 2008–2017 showed that research during the initial 5-year period (2008–2012) followed a similar pattern to Canada, the United Kingdom and the United States, including a preponderance of funding allocated to biomedical research, but this pattern shifted in the latter 5-year period (2013–2017) following the creation of the Autism CRC, resulting in a more even distribution of funding on biological research and topics prioritised by the autistic community, such as services and life span issues (den Houting & Pellicano, 2019; Office of Autism Research Coordination, 2019). This demonstrates that it is possible to respond to the call for a more expansive research agenda led, at least in part, by autistic people themselves.

Conclusion

We are living through a moment of significant change in the way in which people think about autism. That change is reflected in advocacy (e.g. Ne'eman & Bascom, 2020; Rottier & Gernsbacher, 2020), in the media and public debate (e.g. Silberman, 2015), and in arguments about the nature of autism science itself (e.g. Milton, 2014a; Pellicano, 2020; Raymaker, 2020). It is also beginning to be reflected in academic discussions, with a flurry of review papers, perspective pieces and edited books published in the last 12 months alone (Bertilsdotter Rosqvist et al., 2020; Bölte, Lawson, Marschik, & Girdler, 2021; Davis & Crompton, 2021; Kapp, 2020; Leadbitter, Buckle, Ellis, & Dekker, 2021; Mitchell et al., 2021; Walker & Raymaker, 2020). This is a major shift that has revealed fundamental problems in the conventional medical account – and its inability to explain away what Kuhn (1962) called the 'anomalies' at its core – and the possibilities imminent in the neurodiversity paradigm. Some of the commitments at the heart of autism science as we have known it for decades are beginning to wane.

In this review, we have explained that, while the conventional medical paradigm has been a key tool in shaping autism science, it has nevertheless painted an excessively narrow, deficits-based view of autism and excluded the very people it is meant to serve from agenda-setting in research. In so doing, it has placed too many limits on our knowledge of autism and how that knowledge is derived. This is no longer a tenable approach.

We believe the neurodiversity paradigm presents a compelling alternative. It insists that we make a fundamental shift in our most basic assumptions about the nature of autism, broadening and altering our shared theoretical beliefs, values and techniques. Though the neurodiversity paradigm continues to develop and evolve, in its current conception it appears to offer an important means for advancing autism science. As Kuhn explained, paradigm shifts in science are rarely straightforward and, as such, we cannot predict the future shape of autism research with any real degree of accuracy. It is possible that those committed to the conventional medical paradigm will amend their own understanding. It is likely, too, that there will be further developments in our understanding of neurodiversity. Moreover, there will be adherents of both views who will seek to find some kind of mutual accommodation. We already see sign of that third possibility in the World Health Organisation's International Classification of Functioning Disability and Health (ICF), as described in a recent review by Bölte et al. (2021), and in recent developments regarding the importance of optimising the person-environment fit (Lai, Anagnostou, Wiznitzer, Allison, & Baron-Cohen, 2020; Mandy & Lai, 2016).

In among all this uncertainty, however, here we have established three key lessons from the neurodiversity paradigm as currently conceived, which will remain vital for future autism science. First, while the neurodiversity paradigm does not challenge the notion that autism is biological in nature, it stresses the need to view autistic people, not as a collection of 'deficits' needing to be 'fixed' but as unique and worthwhile individuals, whose lives have meaning and purpose. It also urges us to look beyond the individual, focusing on (immediate – especially, relational – and systemic) contexts and the interaction between contextual and individual factors, to address the negative and disabling effects of being autistic. Second, while autism science has a strong record of collaborative efforts between academics (Goldstein, Tager-Flusberg, & Lee, 2015), including autistic researchers and lay members of the autistic community in these collaborations is still a rare occurrence. This needs to change. We need to develop far more robust mechanisms of participatory codesign and coproduction in autism research, to ensure that autism science is designed in partnership with autistic people themselves. Third, once that more participatory spirit does emerge, then we believe autism science will be more likely to focus on autistic community priorities, ensuring that developments in autism science are translated into effective changes in the real-world challenges that autistic people continue to face. That is, after all, why it matters so much in the first place.

做出回应，试图将研究重点导向社区成员优先关注的问题，包括为孤独谱系群体及其家庭识别有效的支持、干预和服务。

澳大利亚的自闭症CRC在这方面的尝试尤为成功。他们采用了全生命周期的孤独谱系研究方法，投资于涵盖早期、学龄期及成年期的项目，并采取了参与式方法，将孤独谱系社群置于其研究工作的核心。这种战略监督取得了成效。对2008-2017年间澳大利亚孤独谱系研究经费支出的分析显示，最初5年（2008-2012年）的研究模式与加拿大、英国和美国相似，包括经费大量分配给生物医学研究，但在自闭症CRC成立后的后5年（2013-2017年），这一模式发生了转变，导致经费在生物学研究以及孤独谱系社群优先考虑的主题（如服务和lifespan问题）上分配更为均衡（den Houting与Pellicano，2019年；孤独症研究协调办公室，2019年）。这表明，响应由孤独谱系群体自身（至少部分地）主导的更广泛研究议程的呼吁是可能的。

结论

我们正经历着人们对孤独谱系的看法发生重大转变的时刻。这种转变体现在倡导活动中（例如Ne'eman与Bascom, 2020；Rottier与Gernsbacher, 2020）、媒体与公众辩论中（例如Silberman, 2015），以及关于孤独谱系科学本质的争论中（例如Milton, 2014a；Pellicano, 2020；Raymaker, 2020）。它也正开始反映在学术讨论中，仅在过去12个月内就涌现了大量综述论文、观点文章和编辑出版的书籍（Bertilsdotter Rosqvist等, 2020；B¨olte, Lawson, Marschik, & Girdler, 2021；Davis与Crompton, 2021；Kapp, 2020；Leadbitter、Buckle、Ellis与Dekker, 2021；Mitchell等, 2021；Walker & Raymaker, 2020）。这是一次重大的转变，揭示了传统医学解释的根本问题——及其无法解释Kuhn（1962）所称的核心"异常现象"——以及神经多样性范式即将带来的可能性。我们数十年来所熟知的孤独谱系科学核心的一些基本信念正开始消退。

在这篇综述中，我们阐述了尽管传统的医学范式一直是塑造孤独谱系科学的关键工具，但它却描绘了一种过于狭隘、基于缺陷的孤独谱系观，并将本应服务于的人群排除在研究议程设定之外。这种做法对我们的认知施加了过多限制。孤独谱系及其知识来源。这已不再是一种可行的途径。

我们认为神经多样性范式提供了一个引人注目的替代方案。它要求我们对孤独谱系本质的最基本假设进行根本性转变，拓宽并改变我们共同的理论信念、价值观和技术。尽管神经多样性范式仍在持续发展和演变，但在其当前的概念中，它似乎为推进孤独谱系科学提供了重要途径。正如Kuhn所解释的，科学中的范式转变很少是直截了当的，因此，我们无法以任何真正的准确性预测孤独谱系研究的未来形态。那些坚持传统医学范式的人可能会修正他们自己的理解。同样，我们对神经多样性的理解也很可能会有进一步发展。此外，两种观点的拥护者都将寻求某种相互调和。我们已经在世界卫生组织的国际功能、残疾和健康分类（ICF）中看到了第三种可能性的迹象，正如B¨olte等人（2021年）最近综述中所描述的，以及在优化人与环境适配重要性方面的近期进展（Lai, Anagnostou, Wiznitzer, Allison, & Baron-Cohen, 2020; Mandy & Lai, 2016）。

然而，在所有这些不确定性之中，我们已从当前所构想的神经多样性范式中确立了三个关键教训，它们对未来孤独谱系科学至关重要。首先，神经多样性范式并不挑战孤独谱系本质上是生物性的观念，但它强调需要将孤独谱系群体视为独特而有价值的个体，其生活具有意义与目的，而非一堆需要"修复"的"缺陷"。它还敦促我们超越个体视角，聚焦于（直接的——尤其是关系性的——以及系统性的）情境以及情境与个体因素之间的互动，以应对身为孤独谱系所带来的负面与致残效应。其次，尽管孤独谱系科学在学者间协作方面有良好记录（Goldstein、Tager-Flusberg 与 Lee，2015），但在这些协作中纳入孤独谱系研究者及孤独谱系社群的普通成员仍属罕见。这需要改变。我们需要在孤独谱系研究中发展更稳健的参与式共同设计与共同生产机制，以确保孤独谱系科学是与孤独谱系群体自身合作设计的。第三，一旦那种更具参与性的精神真正涌现，我们相信孤独谱系科学将更可能聚焦于孤独谱系社群的优先事项，确保孤独谱系科学的进展转化为有效改变，应对孤独谱系群体持续面临的现实世界挑战。毕竟，这正是其如此重要的根本原因。

We are at a turning point in the field of autism science. The conventional medical conceptualisation has played a vital role in autism research and practice to date. Nonetheless, it is increasingly clear that if we wish to make progress for the benefit of the autistic and broader autism communities, it is necessary to re-evaluate our fundamentals. There may well be costs to moving from 'normal' to 'extraordinary' science (Sonuga-Barke, 2020), as well as even more dramatic costs in seeing through the implications of this change. Taking neurodiversity seriously, in other words, would mean both changing the way that we educate the next generation of (autism) scientists, and also demanding substantial real-world changes, overhauling a series of major institutions and practices so that they serve autistic people better. None of this will be easy. Established conventions often remain established long after their original legitimation has dropped away precisely because agreeing to change – funding it, designing it and implanting it – is often extraordinarily difficult (Pierson, 2000). But, crucially, the costs of *not* changing need to be weighed as well. If the experience of other significant moments of social transformation in recent history teaches us anything, it is that previously inconceivable public policy shifts can occur when the demand is strong enough.

Acknowledgements

This work was supported by an Australian Research Council Future Fellowship, awarded to E.P. (FT190100077) and a Macquarie University Research Fellowship awarded to J.d.H. E.P and J.d.H also acknowledge the financial support of the Cooperative Research Centre for Living with Autism (Autism CRC), established and supported under the Australian Government's Cooperative Research Centres Program. The views expressed are the views of the authors alone and do not necessarily represent the views of their organisations or funding sources. The authors are extremely grateful to Marc Stears for critical discussion and for comments on an earlier version of this manuscript. The authors have declared that they have no competing or potential conflicts of interest.

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Key points

- The way in which autism has been conceptualised remains firmly embedded within a conventional medical paradigm.
- Autism is therefore typically understood as a disorder of brain development – an undesirable deviation from the norm.
- This places serious barriers on our understanding of autism and poses deep challenges in autistic people's lives.
- Researchers need, therefore, to consider radically new ways of conceptualising autism.
- The neurodiversity paradigm is one such alternative. It encourages research that is focused on autistic community priorities, and on the interaction between contextual and individual factors, rather than exclusively focused on individual factors.
- Such an approach, which also involves autistic young people and adults in the research process, is crucial to ensuring translational research, with real-world implications for improving autistic lives.

Note

1. In line with the social model of disability and the preferences of the autistic community (Bury, Jellett, Spoor, & Hedley, 2020; Kenny et al., 2016), we use identity-first language (i.e. 'disabled' and 'autistic') in this review.

References

Adams, D., & Young, K. (2020). A systematic review of the perceived barriers and facilitators to accessing psychological treatment for mental health problems in individuals on the autism spectrum. *Review Journal of Autism and Developmental Disorders*, Advance online publication, https://doi.org/10.1007/s40489-020-00226-7

Akhtar, N., & Jaswal, V.K. (2013). Deficit or difference? Interpreting diverse developmental paths: An introduction to the special section. *Developmental Psychology*, *49*, 1–3.

Alvares, G.A., Bebbington, K., Cleary, D., Evans, K., Glasson, E.J., Maybery, M.T., … & Whitehouse, A.J.O. (2020). The misnomer of 'high functioning autism': Intelligence is an imprecise predictor of functional abilities at diagnosis. *Autism*, *24*, 221–232.

Ambitious about Autism (2014). Ruled out: Why are children with autism missing out on education? Available from: https://www.ambitiousaboutautism.org.uk/sites/default/files/resources-and-downloads/files/ruled-out-report-ambitious-about-autism.pdf. Accessed 10th February 2021.

我们正处于孤独谱系科学领域的转折点。传统的医学概念化至今在孤独谱系研究与实践中扮演了关键角色。尽管如此，越来越清晰的是，若我们希望为孤独谱系及更广泛的孤独谱系社群谋取进步，就必须重新评估我们的基本原则。从“常规”科学转向“非凡”科学（Sonuga-Barke，2020）很可能需要付出代价，而看清这一变革的深远影响甚至可能带来更戏剧性的代价。换言之，认真对待神经多样性意味着既要改变我们教育下一代（孤独谱系）科学家的方式，也要要求实质性的现实世界变革，革新一系列主要机构与实践，以更好地服务孤独谱系群体。这一切都不容易。既定惯例往往在其原始正当性早已消失后仍长期存续，正是因为同意变革——资助它、设计它并实施它——通常异常困难（Pierson，2000）。但关键在于，不变革的代价也需要权衡。如果近期历史上其他重大社会转型时刻的经验能给我们任何启示，那就是先前不可想象的公众当需求足够强烈时，政策转变就可能发生。

致谢

本研究由授予E.P.的澳大利亚研究委员会未来研究员基金（FT190100077）和授予J.d.H.的麦考瑞大学研究基金资助。E.P.和J.d.H.亦感谢自闭症生活合作研究中心（Autism CRC）的财务支持，该中心在澳大利亚政府合作研究中心计划下设立并获支持。文中观点仅为作者个人观点，不一定代表其所属机构或资助方的观点。作者对Marc Stears在关键讨论及对本手稿早期版本提出的意见表示衷心感谢。作者声明不存在任何竞争性或潜在的利益冲突。

通讯

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关键要点

- 孤独谱系的概念化方式依然牢固地植根于传统的医学范式之中。
- 因此，孤独谱系通常被理解为一种脑发育障碍——一种偏离常态的不理想状态。
- 这给我们对孤独谱系的理解带来了严重障碍，也给孤独谱系群体的生活造成了深刻挑战。
- 因此，研究人员需要从根本上重新思考如何概念化孤独谱系。
- 神经多样性范式就是这样一种替代方案。它鼓励研究聚焦于孤独谱系社群的优先事项，以及情境因素与个体因素之间的相互作用，而非仅仅专注于个体因素。
- 这样一种方法，在研究过程中也涉及孤独谱系青少年和成年人，对于确保转化研究至关重要，对改善孤独谱系群体的生活具有现实意义。

Note

1. 遵循残障社会模式以及孤独谱系社群的偏好（Bury, Jellett, Spoor, & Hedley, 2020; Kenny等, 2016），我们在本综述中使用身份优先语言（即‘残障’和‘孤独谱系’）。

参考文献

Adams, D., & Young, K. (2020). A systematic review of the perceived barriers and facilitators to accessing psychological treatment for mental health problems in individuals on the autism spectrum. *Review Journal of Autism and Developmental Disorders*, Advance online publication, https://doi.org/10.1007/s40489-020-00226-7

Akhtar, N., & Jaswal, V.K. (2013). Deficit or difference? Interpreting diverse developmental paths: An introduction to the special section. *Developmental Psychology*, *49*, 1–3.

Alvares, G.A., Bebbington, K., Cleary, D., Evans, K., Glasson, E.J., Maybery, M.T., … & Whitehouse, A.J.O. (2020). The misnomer of 'high functioning autism': Intelligence is an imprecise predictor of functional abilities at diagnosis. *Autism*, *24*, 221–232.

Ambitious about Autism (2014). Ruled out: Why are children with autism missing out on education? Available from: https://www.ambitiousaboutautism.org.uk/sites/default/files/resources-and-downloads/files/ruled-out-report-ambitious-about-autism.pdf. Accessed 10th February 2021.

American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th edn). Washington, DC: Author.

Amundson, R. (2000). Against normal function. *Studies in History and Philosophy of Biological and Biomedical Sciences*, *31*, 33–53.

Askari, S., Anaby, D., Bergthorson, M., Majnemer, A., Elsabbagh, M., & Zwaigenbaum, L. (2014). Participation of children and youth with autism spectrum disorder: A scoping review. *Review Journal of Autism and Developmental Disorders*, *2*, 103–114.

Asperger, H. (1944). "Autistic psychopathy" in childhood. In U. Frith (Ed.), *Autism and Asperger syndrome*, U. Frith, Trans. Cambridge, UK: Cambridge University Press.

Bagatell, N. (2010). From cure to community: Transforming notions of autism. *Ethos*, *38*, 33–55.

Ballou, E.P. (2018). What the neurodiversity movement does – and doesn't – offer. *Thinking Person's Guide to Autism*. Available from: http://www.thinkingautismguide.com/2018/02/what-neurodiversity-movement-doesand.html. Accessed 18th July 2021.

Baron-Cohen, S. (2000). Is Asperger syndrome/high-functioning autism necessarily a disability? *Development and Psychopathology*, *12*, 489–500.

Baron-Cohen, S., Leslie, A.M., & Frith, U. (1985). Does the autistic child have a "theory of mind"? *Cognition*, *21*, 37–46.

Bascom, J. (2012). Quiet hands. In The Autistic Self Advocacy Network (Ed.), *Loud hands: Autistic people, speaking* (pp. 177–182). Washington, DC: The Autistic Press.

Becker, H. (1963). *Outsiders*. New York: The Free Press.

Bertilsdotter Rosqvist, H., Chown, N., & Stenning, A. (Eds.) (2020). *Neurodiversity studies: A new critical paradigm*. London: Routledge.

Bertilsdotter Rosqvist, H., & Jackson-Perry, D. (2020). Not doing it properly? (Re)producing and resisting knowledge through narratives of autistic sexualities. *Sexuality and Disability*, Advance online publication.

Blume, H. (1998). Neurodiversity: On the neurological underpinnings of geekdom. *The Atlantic*. Available from: https://archive.is/20130105003900/http://www.theatlantic.com/doc/199809u/neurodiversity. Accessed 10th February 2021.

Bölte, S., Lawson, W.B., Marschik, P.B., & Girdler, S. (2021). Reconciling the seemingly irreconcilable: The WHO's ICF system integrates biological and psychosocial environmental determinants of autism and ADHD. *BioEssays*, *43*, e2000254.

Bottema-Beutel, K., Crowley, S., Sandbank, M., & Woynaroski, T.G. (2021a). Adverse event reporting in intervention research for young autistic children. *Autism*, *25*, 322–335.

Bottema-Beutel, K., Crowley, S., Sandbank, M., & Woynaroski, T.G. (2021b). Research review: Conflicts of Interest (COIs) in autism early intervention research – A meta-analysis of COI influences on intervention effects. *Journal of Child Psychology and Psychiatry*, *62*, 5–15.

Bottema-Beutel, K., Kapp, S.K., Lester, J.N., Sasson, N.J., & Hand, B.N. (2021). Avoiding ableist language: Suggestions for autism researchers. *Autism in Adulthood*, *3*, 18–29.

Brede, J., Remington, A., Kenny, L., Warren, K., & Pellicano, E. (2017). Excluded from school: Autistic students' experiences of school exclusion and subsequent re-integration into school. *Autism & Developmental Language Impairments*, *2*, 1–20. https://doi.org/10.1177/2396941517737511

Brewer, R., Biotti, F., Catmur, C., Press, C., Happé, F., Cook, R., & Bird, G. (2016). Can neurotypical individuals read autistic facial expressions? Atypical production of emotional facial expressions in autism spectrum disorders. *Autism Research*, *9*, 262–271.

Broderick, A.A., & Ne'eman, A. (2008). Autism as metaphor: Narrative and counter-narrative. *International Journal of Inclusive Education*, *12*, 459–476.

Brown-Lavoie, S.M., Viecili, M.A., & Weiss, J.A. (2014). Sexual knowledge and victimization in adults with autism spectrum disorders. *Journal of Autism and Developmental Disorders*, *44*, 2185–2196.

Bury, S.M., Jellett, R., Spoor, J.R., & Hedley, D. (2020). "It defines who I am" or "It's something I have": What language do [autistic] Australian adults [on the autism spectrum] prefer? *Journal of Autism and Developmental Disorders*, Advance online publication. https://doi.org/10.1007/s10803-020-04425-3

Cage, E., Di Monaco, J., & Newell, V. (2018). Experiences of autism acceptance and mental health in autistic adults. *Journal of Autism and Developmental Disorders*, *48*, 473–484.

Cage, E., Di Monaco, J., & Newell, V. (2019). Understanding, attitudes and dehumanisation toward autistic people. *Autism*, *23*, 1373–1383.

Callanan, M., & Waxman, S. (2013). Commentary on special section: Deficit or difference? Interpreting diverse developmental paths. *Developmental Psychology*, *49*, 80–83.

Cassidy, S., Bradley, P., Robinson, J., Allison, C., McHugh, M., & Baron-Cohen, S. (2014). Suicidal ideation and suicide plans or attempts in adults with Asperger's syndrome attending a specialist diagnostic clinic: A clinical cohort study. *The Lancet Psychiatry*, *1*, 142–147.

Chalmers, I. (2004). Well informed uncertainties about the effects of treatments: How should clinicians and patients respond? *BMJ*, *328*, 475–476.

Chen, J.L., Leader, G., Sung, C., & Leahy, M. (2014). Trends in employment for individuals with autism spectrum disorder: A review of the research literature. *Review Journal of Autism and Developmental Disorders*, *2*, 115–127.

Collins, H., & Evans, R. (2002). The third wave of science studies: Studies of expertise and experience. *Social Studies of Science*, *32*, 235–296.

Cooper, K., Smith, L.G.E., & Russell, A. (2017). Social identity, self-esteem, and mental health in autism. *European Journal of Social Psychology*, *47*, 844–854.

Courchesne, V., Girard, D., Jacques, C., & Soulières, I. (2019). Assessing intelligence at autism diagnosis: mission impossible? Testability and cognitive profile of autistic preschoolers. *Journal of Autism and Developmental Disorders*, *49*, 845–856.

Courchesne, V., Meilleur, A.-A.-S., Poulin-Lord, M.-P., Dawson, M., & Soulières, I. (2015). Autistic children at risk of being underestimated: school-based pilot study of a strength-informed assessment. *Molecular Autism*, *6*, 12.

Cowen, T. & Dawson, M. (2018). Michelle Dawson on autism and atypicality. Available from: https://conversationswithtyler.com/episodes/michelle-dawson/. Accessed 15th February 2021.

Cribb, S., Kenny, L., & Pellicano, E. (2019). "I definitely feel more in control of my life": The perspectives of young autistic people and their parents on emerging adulthood. *Autism*, *23*, 1765–1781.

Croen, L.A., Zerbo, O., Qian, Y., Massolo, M.L., Rich, S., Sidney, S., & Kripke, C. (2015). The health status of adults on the autism spectrum. *Autism*, *19*, 814–823.

Crompton, C.J., Ropar, D., Evans-Williams, C.V., Flynn, E.G., & Fletcher-Watson, S. (2020). Autistic peer-to-peer information transfer is highly effective. *Autism*, *24*, 1704–1712.

Crompton, C.J., Sharp, M., Axbey, H., Fletcher-Watson, S., Flynn, E.G., & Ropar, D. (2020). Neurotype-matching, but not being autistic, influences self and observer ratings of interpersonal rapport. *Frontiers in Psychology*, *11*, 586171.

Da Paz, N.S., Siegel, B., Coccia, M.A., & Epel, E.S. (2018). Acceptance or despair? Maternal adjustment to having a child diagnosed with autism. *Journal of Autism and Developmental Disorders*, *48*, 1971–1981.

Davis, R., & Crompton, C.J. (2021). What do new findings about social interaction in autistic adults mean for

American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th edn). Washington, DC: Author.

Amundson, R. (2000). Against normal function. *Studies in History and Philosophy of Biological and Biomedical Sciences*, *31*, 33–53.

Askari, S., Anaby, D., Bergthorson, M., Majnemer, A., Elsabbagh, M., & Zwaigenbaum, L. (2014). Participation of children and youth with autism spectrum disorder: A scoping review. *Review Journal of Autism and Developmental Disorders*, *2*, 103–114.

Asperger, H. (1944). "Autistic psychopathy" in childhood. In U. Frith (Ed.), *Autism and Asperger syndrome*, U. Frith, Trans. Cambridge, UK: Cambridge University Press.

Bagatell, N. (2010). From cure to community: Transforming notions of autism. *Ethos*, *38*, 33–55.

Ballou, E.P. (2018). What the neurodiversity movement does – and doesn't – offer. *Thinking Person's Guide to Autism*. Available from: http://www.thinkingautismguide.com/2018/02/what-neurodiversity-movement-doesand.html. Accessed 18th July 2021.

Baron-Cohen, S. (2000). Is Asperger syndrome/high-functioning autism necessarily a disability? *Development and Psychopathology*, *12*, 489–500.

Baron-Cohen, S., Leslie, A.M., & Frith, U. (1985). Does the autistic child have a "theory of mind"? *Cognition*, *21*, 37–46.

Bascom, J. (2012). Quiet hands. In The Autistic Self Advocacy Network (Ed.), *Loud hands: Autistic people, speaking* (pp. 177–182). Washington, DC: The Autistic Press.

Becker, H. (1963). *Outsiders*. New York: The Free Press.

Bertilsdotter Rosqvist, H., Chown, N., & Stenning, A. (Eds.) (2020). *Neurodiversity studies: A new critical paradigm*. London: Routledge.

Bertilsdotter Rosqvist, H., & Jackson-Perry, D. (2020). Not doing it properly? (Re)producing and resisting knowledge through narratives of autistic sexualities. *Sexuality and Disability*, Advance online publication.

Blume, H. (1998). Neurodiversity: On the neurological underpinnings of geekdom. *The Atlantic*. Available from: https://archive.is/20130105003900/http://www.theatlantic.com/doc/199809u/neurodiversity. Accessed 10th February 2021.

Bölte, S., Lawson, W.B., Marschik, P.B., & Girdler, S. (2021). Reconciling the seemingly irreconcilable: The WHO's ICF system integrates biological and psychosocial environmental determinants of autism and ADHD. *BioEssays*, *43*, e2000254.

Bottema-Beutel, K., Crowley, S., Sandbank, M., & Woynaroski, T.G. (2021a). Adverse event reporting in intervention research for young autistic children. *Autism*, *25*, 322–335.

Bottema-Beutel, K., Crowley, S., Sandbank, M., & Woynaroski, T.G. (2021b). Research review: Conflicts of Interest (COIs) in autism early intervention research – A meta-analysis of COI influences on intervention effects. *Journal of Child Psychology and Psychiatry*, *62*, 5–15.

Bottema-Beutel, K., Kapp, S.K., Lester, J.N., Sasson, N.J., & Hand, B.N. (2021). Avoiding ableist language: Suggestions for autism researchers. *Autism in Adulthood*, *3*, 18–29.

Brede, J., Remington, A., Kenny, L., Warren, K., & Pellicano, E. (2017). Excluded from school: Autistic students' experiences of school exclusion and subsequent re-integration into school. *Autism & Developmental Language Impairments*, *2*, 1–20. https://doi.org/10.1177/2396941517737511

Brewer, R., Biotti, F., Catmur, C., Press, C., Happé, F., Cook, R., & Bird, G. (2016). Can neurotypical individuals read autistic facial expressions? Atypical production of emotional facial expressions in autism spectrum disorders. *Autism Research*, *9*, 262–271.

Broderick, A.A., & Ne'eman, A. (2008). Autism as metaphor: Narrative and counter-narrative. *International Journal of Inclusive Education*, *12*, 459–476.

Brown-Lavoie, S.M., Viecili, M.A., & Weiss, J.A. (2014). Sexual knowledge and victimization in adults with autism spectrum disorders. *Journal of Autism and Developmental Disorders*, *44*, 2185–2196.

Bury, S.M., Jellett, R., Spoor, J.R., & Hedley, D. (2020). "It defines who I am" or "It's something I have": What language do [autistic] Australian adults [on the autism spectrum] prefer? *Journal of Autism and Developmental Disorders*, Advance online publication. https://doi.org/10.1007/s10803-020-04425-3

Cage, E., Di Monaco, J., & Newell, V. (2018). Experiences of autism acceptance and mental health in autistic adults. *Journal of Autism and Developmental Disorders*, *48*, 473–484.

Cage, E., Di Monaco, J., & Newell, V. (2019). Understanding, attitudes and dehumanisation toward autistic people. *Autism*, *23*, 1373–1383.

Callanan, M., & Waxman, S. (2013). Commentary on special section: Deficit or difference? Interpreting diverse developmental paths. *Developmental Psychology*, *49*, 80–83.

Cassidy, S., Bradley, P., Robinson, J., Allison, C., McHugh, M., & Baron-Cohen, S. (2014). Suicidal ideation and suicide plans or attempts in adults with Asperger's syndrome attending a specialist diagnostic clinic: A clinical cohort study. *The Lancet Psychiatry*, *1*, 142–147.

Chalmers, I. (2004). Well informed uncertainties about the effects of treatments: How should clinicians and patients respond? *BMJ*, *328*, 475–476.

Chen, J.L., Leader, G., Sung, C., & Leahy, M. (2014). Trends in employment for individuals with autism spectrum disorder: A review of the research literature. *Review Journal of Autism and Developmental Disorders*, *2*, 115–127.

Collins, H., & Evans, R. (2002). The third wave of science studies: Studies of expertise and experience. *Social Studies of Science*, *32*, 235–296.

Cooper, K., Smith, L.G.E., & Russell, A. (2017). Social identity, self-esteem, and mental health in autism. *European Journal of Social Psychology*, *47*, 844–854.

Courchesne, V., Girard, D., Jacques, C., & Soulières, I. (2019). Assessing intelligence at autism diagnosis: mission impossible? Testability and cognitive profile of autistic preschoolers. *Journal of Autism and Developmental Disorders*, *49*, 845–856.

Courchesne, V., Meilleur, A.-A.-S., Poulin-Lord, M.-P., Dawson, M., & Soulières, I. (2015). Autistic children at risk of being underestimated: school-based pilot study of a strength-informed assessment. *Molecular Autism*, *6*, 12.

Cowen, T. & Dawson, M. (2018). Michelle Dawson on autism and atypicality. Available from: https://conversationswithtyler.com/episodes/michelle-dawson/. Accessed 15th February 2021.

Cribb, S., Kenny, L., & Pellicano, E. (2019). "I definitely feel more in control of my life": The perspectives of young autistic people and their parents on emerging adulthood. *Autism*, *23*, 1765–1781.

Croen, L.A., Zerbo, O., Qian, Y., Massolo, M.L., Rich, S., Sidney, S., & Kripke, C. (2015). The health status of adults on the autism spectrum. *Autism*, *19*, 814–823.

Crompton, C.J., Ropar, D., Evans-Williams, C.V., Flynn, E.G., & Fletcher-Watson, S. (2020). Autistic peer-to-peer information transfer is highly effective. *Autism*, *24*, 1704–1712.

Crompton, C.J., Sharp, M., Axbey, H., Fletcher-Watson, S., Flynn, E.G., & Ropar, D. (2020). Neurotype-matching, but not being autistic, influences self and observer ratings of interpersonal rapport. *Frontiers in Psychology*, *11*, 586171.

Da Paz, N.S., Siegel, B., Coccia, M.A., & Epel, E.S. (2018). Acceptance or despair? Maternal adjustment to having a child diagnosed with autism. *Journal of Autism and Developmental Disorders*, *48*, 1971–1981.

Davis, R., & Crompton, C.J. (2021). What do new findings about social interaction in autistic adults mean for

neurodevelopmental research? *Perspectives on Psychological Science*, *16*, 649–653.
Davis, R., den Houting, J., Nordahl-Hansen, A., & Fletcher-Watson, S. (in press). Helping autistic children. In P. Smith, & C. Hart (Eds.), *The handbook of childhood social development* (3rd edn). Chicester, UK: Wiley-Blackwell.
Dawson, G. (2008). Early behavioral intervention, brain plasticity, and the prevention of autism spectrum disorder. *Development and Psychopathology*, *20*, 775–803.
Dawson, G. (2013). Dramatic increase in autism prevalence parallels explosion of research into its biology and causes. *JAMA Psychiatry*, *70*, 9–10.
Dawson, M. (2004). The misbehaviour of behaviourists: Ethical challenges to the autism-ABA industry. Available from: https://www.sentex.ca/~nexus23/naa_aba.html
Dawson, M., & Fletcher-Watson, S. (2021). Commentary: What conflicts of interest tell us about autism intervention research—A commentary on Bottema-Beutel et al (2020). *Journal of Child Psychology and Psychiatry*, *62*, 16–18.
Dawson, M., & Mottron, L. (2011). Do autistics have cognitive strengths? Should ASC be defined as disorders? In S. Bolte & J. Hallmayer (Eds.), *Autism spectrum conditions: FAQs on autism, Asperger syndrome, and atypical autism answered by international experts* (pp. 32–39). Boston: Hogrefe Publishing.
Dawson, M., Soulieres, I., Gernsbacher, M.A., & Mottron, L. (2007). The level and nature of autistic intelligence. *Psychological Science*, *18*, 657–662.
den Houting, J. (2019). Neurodiversity: An insider's perspective. *Autism*, *23*, 271–273.
den Houting, J., & Pellicano, E. (2019). A portfolio analysis of autism research funding in Australia, 2008–2017. *Journal of Autism and Developmental Disorders*, *49*, 4400–4408.
DiGennaro Reed, F.D., Hirst, J.M., & Hyman, S.R. (2012). Assessment and treatment of stereotypic behavior in children with autism and other developmental disabilities: A thirty year review. *Research in Autism Spectrum Disorders*, *6*, 422–430.
Dinishak, J. (2016). The deficit view and its critics. *Disability Studies Quarterly*, *36*, https://doi.org/10.18061/dsq.v36i4.5236. Accessed 10th February 2021.
Ecker, C., Bookheimer, S.Y., & Murphy, D.G.M. (2015). Neuroimaging in autism spectrum disorder: Brain structure and function across the lifespan. *The Lancet Neurology*, *14*, 1121–1134.
Edey, R., Cook, J., Brewer, R., Johnson, M.H., Bird, G., & Press, C. (2016). Interaction takes two: Typical adults exhibit mind-blindness towards those with autism spectrum disorder. *Journal of Abnormal Psychology*, *125*, 879–885.
Engel, G. (1977). The need for a new medical model: A challenge for biomedicine. *Science*, *196*, 129–136.
Epstein, S. (1996). *Impure science: AIDS, activism, and the politics of knowledge*. Berkeley: University of California Press.
Evans, B. (2013). How autism became autism: The radical transformation of a central concept of child development in Britain. *History of the Human Sciences*, *26*, 3–31.
Fein, D., Barton, M., Eigsti, I.-M., Kelley, E., Naigles, L., Schultz, R.T., … & Tyson, K. (2013). Optimal outcome in individuals with a history of autism. *Journal of Child Psychology and Psychiatry*, *54*, 195–205.
Fenton, A., & Krahn, T. (2007). Autism, neurodiversity and equality beyond the "normal". *Journal of Ethics in Mental Health*, *2*, 1–6.
Fletcher-Watson, S., Adams, J., Brook, K., Charman, T., Crane, L., Cusack, J., … & Pellicano, E. (2019). Making the future together: Shaping autism research through meaningful participation. *Autism*, *23*, 943–953.
Fletcher-Watson, S., & Happé, F. (2019). *Autism: A new introduction to psychological theory and current debate*. London, UK: Routledge.
Frazier, T., Dawson, G., Murray, D., Shih, A., Sachs, J., & Geiger, A. (2018). Brief report: A survey of autism research priorities across a diverse community of stakeholders. *Journal of Autism and Developmental Disorders*, *48*, 3965–3971.
Frith, U., & Happé, F. (1994). Autism: Beyond "theory of mind". *Cognition*, *50*, 115–132.
Frith, U., & Happé, F. (1999). Theory of mind and self-consciousness: What is it like to be autistic? *Mind & Language*, *14*, 1–22.
Gernsbacher, M.A. (2007). Presidential column: The true meaning of research participation. *Association of Psychological Science (APS) Observer*, *20*, 12–13.
Gernsbacher, M.A. (2017). Editorial perspective: The use of person-first language in scholarly writing may accentuate stigma. *Journal of Child Psychology and Psychiatry*, *58*, 859–861.
Gernsbacher, M.A., Dawson, M., & Mottron, L. (2006). Autism: Common, heritable, but not harmful. *Behavioral and Brain Sciences*, *29*, 413–414.
Gernsbacher, M.A., & Yergeau, M. (2019). Empirical failures of the claim that autistic people lack a theory of mind. *Archives of Scientific Psychology*, *7*, 102–118.
Goffman, E. (1990). *Stigma: Notes on the management of spoiled identity* (3rd edn). London: Penguin Books.
Goldstein, N.D., Tager-Flusberg, H., & Lee, B.K. (2015). Mapping collaboration networks in the world of autism research. *Autism Research*, *8*, 1–8.
Grinker, R.R. (2020). Autism, 'stigma', disability: A shifting historical terrain. *Current Anthropology*, *61*, S55–S67.
Happé, F., & Frith, U. (2020). Annual research review: Looking back to look forward – Changes in the concept of autism and implications for future research. *Journal of Child Psychology and Psychiatry*, *61*, 218–232.
Harmuth, E., Silletta, E., Bailey, A., Adams, T., Beck, C., & Barbic, S.P. (2018). Barriers and facilitators to employment for adults with autism: A scoping review. *Annals of International Occupational Therapy*, *1*, 31–40.
Heasman, B., & Gillespie, A. (2019). Neurodivergent intersubjectivity: Distinctive features of how autistic people create shared understanding. *Autism*, *23*, 910–921.
Hedley, D., & Young, R. (2006). Social comparison processes and depressive symptoms in children and adolescents with Asperger syndrome. *Autism*, *10*, 139–153.
Hermelin, B., & Frith, U. (1971). Psychological studies of childhood autism: Can autistic children make sense of what they see and hear? *The Journal of Special Education*, *5*, 107–117.
Hermelin, B., & O'Connor, N. (1975). The recall of digits by normal, deaf and autistic children. *British Journal of Psychology*, *66*, 203–209.
Hickey, G., Richards, T., & Sheehy, J. (2018). Co-production from proposal to paper. *Nature*, *562*, 29–31.
Higgins, J. M., Arnold, S. R., Weise, J. Pellicano, E., & Trollor, J. (2021). Defining autistic burnout through experts by lived experience: Grounded Delphi method investigating# AutisticBurnout. *Autism*, *25*, 2356–2369. https://journals.sagepub.com/doi/abs/10.1177/13623613211019858?journalCode=auta
Hirvikoski, T., Mittendorfer-Rutz, E., Boman, M., Larsson, H., Lichtenstein, P., & Bolte, S. (2016). Premature mortality in autism spectrum disorder. *British Journal of Psychiatry*, *208*, 232–238.
Howlin, P. (2000). Outcome in adult life for more able individuals with autism or Asperger syndrome. *Autism*, *4*, 63–83.
Howlin, P., & Magiati, I. (2017). Autism spectrum disorder: Outcomes in adulthood. *Current Opinions in Psychiatry*, *30*, 69–76.
Hull, L., Petrides, K.V., Allison, C., Smith, P., Baron-Cohen, S., Lai, M.C., & Mandy, W. (2017). "Putting on my best normal": Social camouflaging in adults with autism spectrum

neurodevelopmental research? *Perspectives on Psychological Science*, *16*, 649–653.
Davis, R., den Houting, J., Nordahl-Hansen, A., & Fletcher-Watson, S. (in press). Helping autistic children. In P. Smith, & C. Hart (Eds.), *The handbook of childhood social development* (3rd edn). Chicester, UK: Wiley-Blackwell.
Dawson, G. (2008). Early behavioral intervention, brain plasticity, and the prevention of autism spectrum disorder. *Development and Psychopathology*, *20*, 775–803.
Dawson, G. (2013). Dramatic increase in autism prevalence parallels explosion of research into its biology and causes. *JAMA Psychiatry*, *70*, 9–10.
Dawson, M. (2004). The misbehaviour of behaviourists: Ethical challenges to the autism-ABA industry. Available from: https://www.sentex.ca/~nexus23/naa_aba.html
Dawson, M., & Fletcher-Watson, S. (2021). Commentary: What conflicts of interest tell us about autism intervention research—A commentary on Bottema-Beutel et al (2020). *Journal of Child Psychology and Psychiatry*, *62*, 16–18.
Dawson, M., & Mottron, L. (2011). Do autistics have cognitive strengths? Should ASC be defined as disorders? In S. Bolte & J. Hallmayer (Eds.), *Autism spectrum conditions: FAQs on autism, Asperger syndrome, and atypical autism answered by international experts* (pp. 32–39). Boston: Hogrefe Publishing.
Dawson, M., Soulieres, I., Gernsbacher, M.A., & Mottron, L. (2007). The level and nature of autistic intelligence. *Psychological Science*, *18*, 657–662.
den Houting, J. (2019). Neurodiversity: An insider's perspective. *Autism*, *23*, 271–273.
den Houting, J., & Pellicano, E. (2019). A portfolio analysis of autism research funding in Australia, 2008–2017. *Journal of Autism and Developmental Disorders*, *49*, 4400–4408.
DiGennaro Reed, F.D., Hirst, J.M., & Hyman, S.R. (2012). Assessment and treatment of stereotypic behavior in children with autism and other developmental disabilities: A thirty year review. *Research in Autism Spectrum Disorders*, *6*, 422–430.
Dinishak, J. (2016). The deficit view and its critics. *Disability Studies Quarterly*, *36*, https://doi.org/10.18061/dsq.v36i4.5236. Accessed 10th February 2021.
Ecker, C., Bookheimer, S.Y., & Murphy, D.G.M. (2015). Neuroimaging in autism spectrum disorder: Brain structure and function across the lifespan. *The Lancet Neurology*, *14*, 1121–1134.
Edey, R., Cook, J., Brewer, R., Johnson, M.H., Bird, G., & Press, C. (2016). Interaction takes two: Typical adults exhibit mind-blindness towards those with autism spectrum disorder. *Journal of Abnormal Psychology*, *125*, 879–885.
Engel, G. (1977). The need for a new medical model: A challenge for biomedicine. *Science*, *196*, 129–136.
Epstein, S. (1996). *Impure science: AIDS, activism, and the politics of knowledge*. Berkeley: University of California Press.
Evans, B. (2013). How autism became autism: The radical transformation of a central concept of child development in Britain. *History of the Human Sciences*, *26*, 3–31.
Fein, D., Barton, M., Eigsti, I.-M., Kelley, E., Naigles, L., Schultz, R.T., … & Tyson, K. (2013). Optimal outcome in individuals with a history of autism. *Journal of Child Psychology and Psychiatry*, *54*, 195–205.
Fenton, A., & Krahn, T. (2007). Autism, neurodiversity and equality beyond the "normal". *Journal of Ethics in Mental Health*, *2*, 1–6.
Fletcher-Watson, S., Adams, J., Brook, K., Charman, T., Crane, L., Cusack, J., … & Pellicano, E. (2019). Making the future together: Shaping autism research through meaningful participation. *Autism*, *23*, 943–953.
Fletcher-Watson, S., & Happé, F. (2019). *Autism: A new introduction to psychological theory and current debate*. London, UK: Routledge.
Frazier, T., Dawson, G., Murray, D., Shih, A., Sachs, J., & Geiger, A. (2018). Brief report: A survey of autism research priorities across a diverse community of stakeholders. *Journal of Autism and Developmental Disorders*, *48*, 3965–3971.
Frith, U., & Happé, F. (1994). Autism: Beyond "theory of mind". *Cognition*, *50*, 115–132.
Frith, U., & Happé, F. (1999). Theory of mind and self-consciousness: What is it like to be autistic? *Mind & Language*, *14*, 1–22.
Gernsbacher, M.A. (2007). Presidential column: The true meaning of research participation. *Association of Psychological Science (APS) Observer*, *20*, 12–13.
Gernsbacher, M.A. (2017). Editorial perspective: The use of person-first language in scholarly writing may accentuate stigma. *Journal of Child Psychology and Psychiatry*, *58*, 859–861.
Gernsbacher, M.A., Dawson, M., & Mottron, L. (2006). Autism: Common, heritable, but not harmful. *Behavioral and Brain Sciences*, *29*, 413–414.
Gernsbacher, M.A., & Yergeau, M. (2019). Empirical failures of the claim that autistic people lack a theory of mind. *Archives of Scientific Psychology*, *7*, 102–118.
Goffman, E. (1990). *Stigma: Notes on the management of spoiled identity* (3rd edn). London: Penguin Books.
Goldstein, N.D., Tager-Flusberg, H., & Lee, B.K. (2015). Mapping collaboration networks in the world of autism research. *Autism Research*, *8*, 1–8.
Grinker, R.R. (2020). Autism, 'stigma', disability: A shifting historical terrain. *Current Anthropology*, *61*, S55–S67.
Happé, F., & Frith, U. (2020). Annual research review: Looking back to look forward – Changes in the concept of autism and implications for future research. *Journal of Child Psychology and Psychiatry*, *61*, 218–232.
Harmuth, E., Silletta, E., Bailey, A., Adams, T., Beck, C., & Barbic, S.P. (2018). Barriers and facilitators to employment for adults with autism: A scoping review. *Annals of International Occupational Therapy*, *1*, 31–40.
Heasman, B., & Gillespie, A. (2019). Neurodivergent intersubjectivity: Distinctive features of how autistic people create shared understanding. *Autism*, *23*, 910–921.
Hedley, D., & Young, R. (2006). Social comparison processes and depressive symptoms in children and adolescents with Asperger syndrome. *Autism*, *10*, 139–153.
Hermelin, B., & Frith, U. (1971). Psychological studies of childhood autism: Can autistic children make sense of what they see and hear? *The Journal of Special Education*, *5*, 107–117.
Hermelin, B., & O'Connor, N. (1975). The recall of digits by normal, deaf and autistic children. *British Journal of Psychology*, *66*, 203–209.
Hickey, G., Richards, T., & Sheehy, J. (2018). Co-production from proposal to paper. *Nature*, *562*, 29–31.
Higgins, J. M., Arnold, S. R., Weise, J. Pellicano, E., & Trollor, J. (2021). Defining autistic burnout through experts by lived experience: Grounded Delphi method investigating# AutisticBurnout. *Autism*, *25*, 2356–2369. https://journals.sagepub.com/doi/abs/10.1177/13623613211019858?journalCode=auta
Hirvikoski, T., Mittendorfer-Rutz, E., Boman, M., Larsson, H., Lichtenstein, P., & Bolte, S. (2016). Premature mortality in autism spectrum disorder. *British Journal of Psychiatry*, *208*, 232–238.
Howlin, P. (2000). Outcome in adult life for more able individuals with autism or Asperger syndrome. *Autism*, *4*, 63–83.
Howlin, P., & Magiati, I. (2017). Autism spectrum disorder: Outcomes in adulthood. *Current Opinions in Psychiatry*, *30*, 69–76.
Hull, L., Petrides, K.V., Allison, C., Smith, P., Baron-Cohen, S., Lai, M.C., & Mandy, W. (2017). "Putting on my best normal": Social camouflaging in adults with autism spectrum

conditions. *Journal of Autism and Developmental Disorders*, *47*, 2519–2534.
Humphrey, N., & Hebron, J. (2014). Bullying of children and adolescents with autism spectrum conditions: A 'state of the field' review. *International Journal of Inclusive Education*, *19*, 845–862.
Humphrey, N., & Lewis, S. (2008). "Make me normal": The views and experiences of pupils on the autistic spectrum in mainstream secondary schools. *Autism*, *12*, 23–46.
Insel, T., & Daniels, S. (2011). Future directions: Setting priorities to guide the federal research effort. In D. Amaral, G. Dawson & D. Geschwind (Eds.), *Autism spectrum disorders* (pp. 1359–1366). New York: Oxford University Press.
Jaarsma, P., & Welin, S. (2012). Autism as a natural human variation: Reflections on the claims of the neurodiversity movement. *Health Care Analysis*, *20*, 20–30.
Jaswal, V.K., & Akhtar, N. (2018). Being vs. Appearing socially uninterested: Challenging assumptions about social motivation in autism. *Behavioral and Brain Sciences*, *42*, 1–73.
Jones, E.J., Gliga, T., Bedford, R., Charman, T., & Johnson, M.H. (2014). Developmental pathways to autism: A review of prospective studies of infants at risk. *Neuroscience and Biobehavioral Reviews*, *39*, 1–33.
Jose, C., George-Zwicker, P., Tardif, L., Bouma, A., Pugsley, D., Pugsley, L., … & Robichaud, M. (2020). "We are the stakeholders with the most at stake": Scientific and autism community co-researchers reflect on their collaborative experience in the CONNECT project. *Research Involvement and Engagement*, *6*, 58.
Joyce, C., Honey, E., Leekam, S.R., Barrett, S.L., & Rodgers, J. (2017). Anxiety, intolerance of uncertainty and restricted and repetitive behaviour: Insights directly from young people with ASD. *Journal of Autism and Developmental Disorders*, *47*, 3789–3802.
Jumah, F., Ghannam, M., Jaber, M., Adeeb, N., & Tubbs, R.S. (2016). Neuroanatomical variation in autism spectrum disorder: A comprehensive review. *Clinical Anatomy*, *29*, 454–465.
Kanner, L. (1943). Autistic disturbances of affective contact. *Pathology*, 217–250.
Kapp, S. (Ed.) (2020). *Autistic community and the neurodiversity movement: Stories from the frontline*. Singapore: Palgrave Macmillan.
Kapp, S., Steward, R., Crane, L., Elliott, D., Elphick, C., Pellicano, E., & Russell, G. (2019). "People should be allowed to do what they like": Autistic adults' views and experiences of stimming. *Autism*, *23*, 1782–1792.
Kenny, L., Hattersley, C., Molins, B., Buckley, C., Povey, C., & Pellicano, E. (2016). Which terms should be used to describe autism? Perspectives from the UK autism community. *Autism*, *20*, 442–462.
Kirby, A.V., Bakian, A.V., Zhang, Y., Bilder, D.A., Keeshin, B.R., & Coon, H. (2019). A 20-year study of suicide death in a statewide autism population. *Autism Research*, *12*, 658–666.
Kittay, E. (2017). The moral significance of being human. *Proceedings and Addresses of the American Philosophical Association*, *91*, 22–42.
Knapp, M., Romeo, R., & Beecham, J. (2009). Economic cost of autism in the UK. *Autism*, *13*, 317–336.
Kolevzon, A., Gross, R., & Reichenberg, A. (2007). Prenatal and perinatal risk factors for autism: A review and integration of findings. *Archives of Pediatric and Adolescent Medicine*, *161*, 326–333.
Kuhn, T. (1962). Historical structure of scientific discovery. *Science*, *136*, 760–764.
Lai, M.-C., Anagnostou, E., Wiznitzer, M., Allison, C., & Baron-Cohen, S. (2020). Evidence-based support for autistic people across the lifespan: Maximising potential, minimising barriers, and optimising the person-environment fit. *Lancet Neurology*, *19*, 434–451.
Lai, M.-C., Kassee, C., Besney, R., Bonato, S., Hull, L., Mandy, W., … & Ameis, S.H. (2019). Prevalence of co-occurring mental health diagnoses in the autism population: A systematic review and meta-analysis. *The Lancet Psychiatry*, *6*, 819–829.
Lam, G.Y.H., Sabnis, S., Migueliz Valcarlos, M., & Wolgemuth, J.R. (2021). A critical review of academic literature constructing well-being in autistic adults. *Autism in Adulthood*, *3*, 61–71.
Leadbitter, K., Buckle, K.L., Ellis, C., & Dekker, M. (2021). Autistic self-advocacy and the neurodiversity movement: Implications for autism early intervention research and practice. *Frontiers in Psychology*, *12*(Article 635690), 1–7. https://doi.org/10.3389/fpsyg.2021.635690
Leigh, J.P., & Du, J. (2015). Brief report: Forecasting the economic burden of autism in 2015 and 2025 in the United States. *Journal of Autism and Developmental Disorders*, *45*, 4135–4139.
Lilley, R. (2018). *What's in a flap? The curious history of autism and hand stereotypies*. Paper presented at the Neurosocieties Symposium: Explorations of the brain, culture, and ethics, Monash University.
Llewellyn, A., & Hogan, K. (2000). The use and abuse of models of disability. *Disability & Society*, *15*, 157–165.
Lloyd, K., & White, J. (2011). Democratizing clinical research. *Nature*, *474*, 277–278.
Lombardo, M.V., Chakrabarti, B., Bullmore, E.T., MRC AIMS Consortium, & Baron-Cohen, S. (2011). Specialization of right temporo-parietal junction for mentalizing and its relation to social impairments in autism. *NeuroImage*, *56*, 1832–1838.
Lory, B. (2019). For the love of science? *Autism in Adulthood*, *1*, 12–14.
Lovaas, O.I. (1987). Behavioral treatment and normal educational and intellectual functioning in young autistic children. *Journal of Consulting and Clinical Psychology*, *55*, 3–9.
Maiano, C., Normand, C.L., Salvas, M.C., Moullec, G., & Aime, A. (2016). Prevalence of school bullying among youth with autism spectrum disorders: A systematic review and meta-analysis. *Autism Research*, *9*, 601–615.
Mandy, W., & Lai, M.-C. (2016). Annual Research Review: The role of the environment in the developmental psychopathology of autism spectrum condition. *Journal of Child Psychology and Psychiatry*, *57*, 271–292.
Mandy, W., Murin, M., Baykaner, O., Staunton, S., Cobb, R., Hellriegel, J., … & Skuse, D. (2016). Easing the transition to secondary education for children with autism spectrum disorder: An evaluation of the Systemic Transition in Education Programme for Autism Spectrum Disorder (STEP-ASD). *Autism*, *20*, 580–590.
Marks, D. (1997). Models of disability. *Disability and Rehabilitation*, *19*, 85–91.
Marsack-Topolewski, C.N., Samuel, P.S., & Tarraf, W. (2021). Empirical evaluation of the association between daily living skills of adults with autism and parental caregiver burden. *PLoS One*, *16*, e0244844.
Mason, D., Ingham, B., Urbanowicz, A., Michael, C., Birtles, H., Woodbury-Smith, M., … & Parr, J.R. (2019). A systematic review of what barriers and facilitators prevent and enable physical healthcare services access for autistic adults. *Journal of Autism and Developmental Disorders*, *49*, 3387–3400.
Mazefsky, C.A., Kao, J., & Oswald, D.P. (2011). Preliminary evidence suggesting caution in the use of psychiatric self-report measures with adolescents with high-functioning autism spectrum disorders. *Research in Autism Spectrum Disorders*, *5*, 164–174.
McConachie, H., Wilson, C., Mason, D., Garland, D., Parr, J.R., Rattazzi, A., … & Magiati, I. (2020). What is important in measuring quality of life? Reflections by autistic adults in four countries. *Autism in Adulthood*, *2*, 4–12.

conditions. *Journal of Autism and Developmental Disorders*, *47*, 2519–2534.
Humphrey, N., & Hebron, J. (2014). Bullying of children and adolescents with autism spectrum conditions: A 'state of the field' review. *International Journal of Inclusive Education*, *19*, 845–862.
Humphrey, N., & Lewis, S. (2008). "Make me normal": The views and experiences of pupils on the autistic spectrum in mainstream secondary schools. *Autism*, *12*, 23–46.
Insel, T., & Daniels, S. (2011). Future directions: Setting priorities to guide the federal research effort. In D. Amaral, G. Dawson & D. Geschwind (Eds.), *Autism spectrum disorders* (pp. 1359–1366). New York: Oxford University Press.
Jaarsma, P., & Welin, S. (2012). Autism as a natural human variation: Reflections on the claims of the neurodiversity movement. *Health Care Analysis*, *20*, 20–30.
Jaswal, V.K., & Akhtar, N. (2018). Being vs. Appearing socially uninterested: Challenging assumptions about social motivation in autism. *Behavioral and Brain Sciences*, *42*, 1–73.
Jones, E.J., Gliga, T., Bedford, R., Charman, T., & Johnson, M.H. (2014). Developmental pathways to autism: A review of prospective studies of infants at risk. *Neuroscience and Biobehavioral Reviews*, *39*, 1–33.
Jose, C., George-Zwicker, P., Tardif, L., Bouma, A., Pugsley, D., Pugsley, L., … & Robichaud, M. (2020). "We are the stakeholders with the most at stake": Scientific and autism community co-researchers reflect on their collaborative experience in the CONNECT project. *Research Involvement and Engagement*, *6*, 58.
Joyce, C., Honey, E., Leekam, S.R., Barrett, S.L., & Rodgers, J. (2017). Anxiety, intolerance of uncertainty and restricted and repetitive behaviour: Insights directly from young people with ASD. *Journal of Autism and Developmental Disorders*, *47*, 3789–3802.
Jumah, F., Ghannam, M., Jaber, M., Adeeb, N., & Tubbs, R.S. (2016). Neuroanatomical variation in autism spectrum disorder: A comprehensive review. *Clinical Anatomy*, *29*, 454–465.
Kanner, L. (1943). Autistic disturbances of affective contact. *Pathology*, 217–250.
Kapp, S. (Ed.) (2020). *Autistic community and the neurodiversity movement: Stories from the frontline*. Singapore: Palgrave Macmillan.
Kapp, S., Steward, R., Crane, L., Elliott, D., Elphick, C., Pellicano, E., & Russell, G. (2019). "People should be allowed to do what they like": Autistic adults' views and experiences of stimming. *Autism*, *23*, 1782–1792.
Kenny, L., Hattersley, C., Molins, B., Buckley, C., Povey, C., & Pellicano, E. (2016). Which terms should be used to describe autism? Perspectives from the UK autism community. *Autism*, *20*, 442–462.
Kirby, A.V., Bakian, A.V., Zhang, Y., Bilder, D.A., Keeshin, B.R., & Coon, H. (2019). A 20-year study of suicide death in a statewide autism population. *Autism Research*, *12*, 658–666.
Kittay, E. (2017). The moral significance of being human. *Proceedings and Addresses of the American Philosophical Association*, *91*, 22–42.
Knapp, M., Romeo, R., & Beecham, J. (2009). Economic cost of autism in the UK. *Autism*, *13*, 317–336.
Kolevzon, A., Gross, R., & Reichenberg, A. (2007). Prenatal and perinatal risk factors for autism: A review and integration of findings. *Archives of Pediatric and Adolescent Medicine*, *161*, 326–333.
Kuhn, T. (1962). Historical structure of scientific discovery. *Science*, *136*, 760–764.
Lai, M.-C., Anagnostou, E., Wiznitzer, M., Allison, C., & Baron-Cohen, S. (2020). Evidence-based support for autistic people across the lifespan: Maximising potential, minimising barriers, and optimising the person-environment fit. *Lancet Neurology*, *19*, 434–451.
Lai, M.-C., Kassee, C., Besney, R., Bonato, S., Hull, L., Mandy, W., … & Ameis, S.H. (2019). Prevalence of co-occurring mental health diagnoses in the autism population: A systematic review and meta-analysis. *The Lancet Psychiatry*, *6*, 819–829.
Lam, G.Y.H., Sabnis, S., Migueliz Valcarlos, M., & Wolgemuth, J.R. (2021). A critical review of academic literature constructing well-being in autistic adults. *Autism in Adulthood*, *3*, 61–71.
Leadbitter, K., Buckle, K.L., Ellis, C., & Dekker, M. (2021). Autistic self-advocacy and the neurodiversity movement: Implications for autism early intervention research and practice. *Frontiers in Psychology*, *12*(Article 635690), 1–7. https://doi.org/10.3389/fpsyg.2021.635690
Leigh, J.P., & Du, J. (2015). Brief report: Forecasting the economic burden of autism in 2015 and 2025 in the United States. *Journal of Autism and Developmental Disorders*, *45*, 4135–4139.
Lilley, R. (2018). *What's in a flap? The curious history of autism and hand stereotypies*. Paper presented at the Neurosocieties Symposium: Explorations of the brain, culture, and ethics, Monash University.
Llewellyn, A., & Hogan, K. (2000). The use and abuse of models of disability. *Disability & Society*, *15*, 157–165.
Lloyd, K., & White, J. (2011). Democratizing clinical research. *Nature*, *474*, 277–278.
Lombardo, M.V., Chakrabarti, B., Bullmore, E.T., MRC AIMS Consortium, & Baron-Cohen, S. (2011). Specialization of right temporo-parietal junction for mentalizing and its relation to social impairments in autism. *NeuroImage*, *56*, 1832–1838.
Lory, B. (2019). For the love of science? *Autism in Adulthood*, *1*, 12–14.
Lovaas, O.I. (1987). Behavioral treatment and normal educational and intellectual functioning in young autistic children. *Journal of Consulting and Clinical Psychology*, *55*, 3–9.
Maiano, C., Normand, C.L., Salvas, M.C., Moullec, G., & Aime, A. (2016). Prevalence of school bullying among youth with autism spectrum disorders: A systematic review and meta-analysis. *Autism Research*, *9*, 601–615.
Mandy, W., & Lai, M.-C. (2016). Annual Research Review: The role of the environment in the developmental psychopathology of autism spectrum condition. *Journal of Child Psychology and Psychiatry*, *57*, 271–292.
Mandy, W., Murin, M., Baykaner, O., Staunton, S., Cobb, R., Hellriegel, J., … & Skuse, D. (2016). Easing the transition to secondary education for children with autism spectrum disorder: An evaluation of the Systemic Transition in Education Programme for Autism Spectrum Disorder (STEP-ASD). *Autism*, *20*, 580–590.
Marks, D. (1997). Models of disability. *Disability and Rehabilitation*, *19*, 85–91.
Marsack-Topolewski, C.N., Samuel, P.S., & Tarraf, W. (2021). Empirical evaluation of the association between daily living skills of adults with autism and parental caregiver burden. *PLoS One*, *16*, e0244844.
Mason, D., Ingham, B., Urbanowicz, A., Michael, C., Birtles, H., Woodbury-Smith, M., … & Parr, J.R. (2019). A systematic review of what barriers and facilitators prevent and enable physical healthcare services access for autistic adults. *Journal of Autism and Developmental Disorders*, *49*, 3387–3400.
Mazefsky, C.A., Kao, J., & Oswald, D.P. (2011). Preliminary evidence suggesting caution in the use of psychiatric self-report measures with adolescents with high-functioning autism spectrum disorders. *Research in Autism Spectrum Disorders*, *5*, 164–174.
McConachie, H., Wilson, C., Mason, D., Garland, D., Parr, J.R., Rattazzi, A., … & Magiati, I. (2020). What is important in measuring quality of life? Reflections by autistic adults in four countries. *Autism in Adulthood*, *2*, 4–12.

McGeer, V. (2004). Autistic self-awareness. *Philosophy, Psychiatry, & Psychology*, *11*, 235–251.
Medin, D., Bennis, W., & Chandler, M. (2010). Culture and the home-field disadvantage. *Perspectives on Psychological Science*, *5*, 708–713.
Milton, D. (2012). On the ontological status of autism: The 'double empathy problem'. *Disability & Society*, *27*, 883–887.
Milton, D. (2014a). Autistic expertise: A critical reflection on the production of knowledge in autism studies. *Autism*, *18*, 794–802.
Milton, D. (2014b). So what exactly are autism interventions intervening with? *Good Autism Practice*, *15*, 6–14.
Milton, D., & Bracher, M. (2013). Autistics speak but are they heard? *Medical Sociology Online*, *7*, 61–69.
Milton, D., Heasman, B., & Sheppard, E. (2018). Double empathy. In F.R. Volkmar (Ed.), *Encyclopedia of autism spectrum disorders* (pp. 1–8). New York: Springer.
Mitchell, P., Sheppard, E., & Cassidy, S. (2021). Autism and the double empathy problem: Implications for development and mental health. *British Journal of Developmental Psychology*, *39*, 1–18.
Modabbernia, A., Velthorst, E., & Reichenberg, A. (2017). Environmental risk factors for autism: An evidence-based review of systematic reviews and meta-analyses. *Molecular Autism*, *8*, 13.
Moore, K. (2008). *Disrupting science: Social movements, American scientists, and the politics of the military, 1945–1975.* Princeton, NJ: Princeton University Press.
Morrison, K.E., DeBrabander, K.M., Faso, D.J., & Sasson, N.J. (2019). Variability in first impressions of autistic adults made by neurotypical raters is driven more by characteristics of the rater than by characteristics of autistic adults. *Autism*, *23*, 1817–1829.
Morrison, K.E., DeBrabander, K.M., Jones, D.R., Faso, D.J., Ackerman, R.A., & Sasson, N.J. (2020). Outcomes of real-world social interaction for autistic adults paired with autistic compared to typically developing partners. *Autism*, *24*, 1067–1080.
Mottron, L. (2011). The power of autism. *Nature*, *479*, 33–35.
Mottron, L., & Belleville, S. (1993). A study of perceptual analysis in a high-level autistic subject with exceptional graphic abilities. *Brain and Cognition*, *23*, 279–309.
Muth, A., Honekopp, J., & Falter, C.M. (2014). Visuo-spatial performance in autism: A meta-analysis. *Journal of Autism and Developmental Disorders*, *44*, 3245–3263.
National Health and Medical Research Council. (1991). *Guidelines on ethical matters in Aboriginal and Torres Strait Islander research.* Canberra: Commonwealth of Australia.
Ne'eman, A. (2021). When disability is defined by behavior, outcome measures should not promote 'passing'. *AMA Journal of Ethics*, *23*, E569–E575.
Ne'eman, A., & Bascom, J. (2020). Autistic self-advocacy in the developmental disability movement. *The American Journal of Bioethics*, *20*, 25–27.
Neumeier, S., & Brown, L. (2020). Torture in the name of treatment: The mission to stop the shocks in the age of deinstitutionalisation. In S.K. Kapp (Ed.), *Autistic community and the neurodiversity movement: Stories from the frontline* (pp. 195–210). Singapore: Palgrave Macmillan.
Nicolaidis, C. (2012). What can physicians learn from the neurodiversity movement? *American Medical Association Journal of Ethics*, *14*, 503–510.
Nicolaidis, C., Raymaker, D.M., Ashkenazy, E., McDonald, K.E., Dern, S., Baggs, A.E.V., … & Boisclair, W.C. (2015). "Respect the way I need to communicate with you": Healthcare experiences of adults on the autism spectrum. *Autism*, *19*, 824–831.
Nicolaidis, C., Raymaker, D., McDonald, K., Dern, S., Ashkenazy, E., Boisclair, C., … & Baggs, A. (2011). Collaboration strategies in nontraditional community-based participatory research partnerships: Lessons from an academic-community partnership with autistic self-advocates. *Progress in Community Health Partnerships: Research, Education, and Action*, *5*, 143–150.
Office of Autism Research Coordination (NIMH) on behalf of the Interagency Autism Coordinating Committee (IACC) (2012). 2010 Autism spectrum disorder research: Portfolio analysis report. Available from: https://iacc.hhs.gov/publications/portfolio-analysis/2010/. Accessed 18th February 2021.
Office of Autism Research Coordination (NIMH) on behalf of the Interagency Autism Coordinating Committee (IACC) (2017). Portfolio analysis report IACC Autism spectrum disorder research 2014–2015. Available from: https://iacc.hhs.gov/publications/portfolio-analysis/2015/. Accessed 18th February 2021.
Office of Autism Research Coordination (NIMH), Autistica, Canadian Institutes of Health Research, and Macquarie University (2019). 2016 International Autism Spectrum Disorder Research Portfolio Analysis Report. Available from: https://iacc.hhs.gov/publications/international-portfolio-analysis/2016/portfolio_analysis_2016.pdf. Accessed 6th February 2021.
Oliver, M. (1996). *Understanding disability: From theory to practice.* Chatham, UK: Mackays of Chatham.
Orinstein, A.J., Helt, M., Troyb, E., Tyson, K.E., Barton, M.L., Eigsti, I.-M., … & Fein, D.A. (2014). Intervention for optimal outcome in children and adolescents with a history of autism. *Journal of Developmental and Behavioral Pediatrics*, *35*, 247–256.
Pearson, A., & Rose, K. (2021). A conceptual analysis of autistic masking: Understanding the narrative of stigma and the illusion of choice. *Autism in Adulthood*, *3*, 52–60.
Pellicano, E. (2020). Commentary: Broadening the research remit of participatory methods in autism science – a commentary on Happé and Frith (2020). *Journal of Child Psychology and Psychiatry*, *61*, 233–235.
Pellicano, E., & Burr, D. (2012). When the world becomes 'too real': A Bayesian explanation of autistic perception. *Trends in Cognitive Sciences*, *16*, 504–510.
Pellicano, E., Dinsmore, A. & Charman, T. (2013). A future made together: Shaping autism research in the UK. Available from: http://discovery.ucl.ac.uk/10017703/1/A_Future_Made_Together_1.2_LR.pdf. Accessed 20th February 2021
Pellicano, E., Dinsmore, A., & Charman, T. (2014a). Views on researcher-community engagement in autism research in the United Kingdom: A mixed-methods study. *PLoS One*, *9*, 1–11.
Pellicano, E., Dinsmore, A., & Charman, T. (2014b). What should autism research focus upon? Community views and priorities from the United Kingdom. *Autism*, *18*, 756–770.
Pellicano, E., Jeffery, L., Burr, D., & Rhodes, G. (2007). Abnormal adaptive face-coding mechanisms in children with autism spectrum disorder. *Current Biology*, *17*, 1508–1512.
Pellicano, E., & Stears, M. (2011). Bridging autism, science and society: Moving toward an ethically informed approach to autism research. *Autism Research*, *4*, 271–282.
Picardi, A., Gigantesco, A., Tarolla, E., Stoppioni, V., Cerbo, R., Cremonte, M., … & Nardocci, F. (2018). Parental burden and its correlates in families of children with autism spectrum disorder: A multicentre study with two comparison groups. *Clinical Practice & Epidemiology in Mental Health*, *14*, 143–176.
Pierson, P. (2000). Increasing returns, path dependence, and the study of politics. *American Political Science Review, 94*, 251–267.
Pripas-Kapit, S. (2020). Historicizing Jim Sinclair's "Don't Mourn for Us": A cultural and intellectual history of neurodiversity's first manifesto. In S.K. Kapp (Ed.), *Autistic community and the neurodiversity movement: Stories from the frontline* (pp. 23–39). Singapore: Palgrave Macmillan.

McGeer, V. (2004). Autistic self-awareness. *Philosophy, Psychiatry, & Psychology*, *11*, 235–251.
Medin, D., Bennis, W., & Chandler, M. (2010). Culture and the home-field disadvantage. *Perspectives on Psychological Science*, *5*, 708–713.
Milton, D. (2012). On the ontological status of autism: The 'double empathy problem'. *Disability & Society*, *27*, 883–887.
Milton, D. (2014a). Autistic expertise: A critical reflection on the production of knowledge in autism studies. *Autism*, *18*, 794–802.
Milton, D. (2014b). So what exactly are autism interventions intervening with? *Good Autism Practice*, *15*, 6–14.
Milton, D., & Bracher, M. (2013). Autistics speak but are they heard? *Medical Sociology Online*, *7*, 61–69.
Milton, D., Heasman, B., & Sheppard, E. (2018). Double empathy. In F.R. Volkmar (Ed.), *Encyclopedia of autism spectrum disorders* (pp. 1–8). New York: Springer.
Mitchell, P., Sheppard, E., & Cassidy, S. (2021). Autism and the double empathy problem: Implications for development and mental health. *British Journal of Developmental Psychology*, *39*, 1–18.
Modabbernia, A., Velthorst, E., & Reichenberg, A. (2017). Environmental risk factors for autism: An evidence-based review of systematic reviews and meta-analyses. *Molecular Autism*, *8*, 13.
Moore, K. (2008). *Disrupting science: Social movements, American scientists, and the politics of the military, 1945–1975.* Princeton, NJ: Princeton University Press.
Morrison, K.E., DeBrabander, K.M., Faso, D.J., & Sasson, N.J. (2019). Variability in first impressions of autistic adults made by neurotypical raters is driven more by characteristics of the rater than by characteristics of autistic adults. *Autism*, *23*, 1817–1829.
Morrison, K.E., DeBrabander, K.M., Jones, D.R., Faso, D.J., Ackerman, R.A., & Sasson, N.J. (2020). Outcomes of real-world social interaction for autistic adults paired with autistic compared to typically developing partners. *Autism*, *24*, 1067–1080.
Mottron, L. (2011). The power of autism. *Nature*, *479*, 33–35.
Mottron, L., & Belleville, S. (1993). A study of perceptual analysis in a high-level autistic subject with exceptional graphic abilities. *Brain and Cognition*, *23*, 279–309.
Muth, A., Honekopp, J., & Falter, C.M. (2014). Visuo-spatial performance in autism: A meta-analysis. *Journal of Autism and Developmental Disorders*, *44*, 3245–3263.
National Health and Medical Research Council. (1991). *Guidelines on ethical matters in Aboriginal and Torres Strait Islander research.* Canberra: Commonwealth of Australia.
Ne'eman, A. (2021). When disability is defined by behavior, outcome measures should not promote 'passing'. *AMA Journal of Ethics*, *23*, E569–E575.
Ne'eman, A., & Bascom, J. (2020). Autistic self-advocacy in the developmental disability movement. *The American Journal of Bioethics*, *20*, 25–27.
Neumeier, S., & Brown, L. (2020). Torture in the name of treatment: The mission to stop the shocks in the age of deinstitutionalisation. In S.K. Kapp (Ed.), *Autistic community and the neurodiversity movement: Stories from the frontline* (pp. 195–210). Singapore: Palgrave Macmillan.
Nicolaidis, C. (2012). What can physicians learn from the neurodiversity movement? *American Medical Association Journal of Ethics*, *14*, 503–510.
Nicolaidis, C., Raymaker, D.M., Ashkenazy, E., McDonald, K.E., Dern, S., Baggs, A.E.V., … & Boisclair, W.C. (2015). "Respect the way I need to communicate with you": Healthcare experiences of adults on the autism spectrum. *Autism*, *19*, 824–831.
Nicolaidis, C., Raymaker, D., McDonald, K., Dern, S., Ashkenazy, E., Boisclair, C., … & Baggs, A. (2011). Collaboration strategies in nontraditional community-based participatory research partnerships: Lessons from an academic-community partnership with autistic self-advocates. *Progress in Community Health Partnerships: Research, Education, and Action*, *5*, 143–150.
Office of Autism Research Coordination (NIMH) on behalf of the Interagency Autism Coordinating Committee (IACC) (2012). 2010 Autism spectrum disorder research: Portfolio analysis report. Available from: https://iacc.hhs.gov/publications/portfolio-analysis/2010/. Accessed 18th February 2021.
Office of Autism Research Coordination (NIMH) on behalf of the Interagency Autism Coordinating Committee (IACC) (2017). Portfolio analysis report IACC Autism spectrum disorder research 2014–2015. Available from: https://iacc.hhs.gov/publications/portfolio-analysis/2015/. Accessed 18th February 2021.
Office of Autism Research Coordination (NIMH), Autistica, Canadian Institutes of Health Research, and Macquarie University (2019). 2016 International Autism Spectrum Disorder Research Portfolio Analysis Report. Available from: https://iacc.hhs.gov/publications/international-portfolio-analysis/2016/portfolio_analysis_2016.pdf. Accessed 6th February 2021.
Oliver, M. (1996). *Understanding disability: From theory to practice.* Chatham, UK: Mackays of Chatham.
Orinstein, A.J., Helt, M., Troyb, E., Tyson, K.E., Barton, M.L., Eigsti, I.-M., … & Fein, D.A. (2014). Intervention for optimal outcome in children and adolescents with a history of autism. *Journal of Developmental and Behavioral Pediatrics*, *35*, 247–256.
Pearson, A., & Rose, K. (2021). A conceptual analysis of autistic masking: Understanding the narrative of stigma and the illusion of choice. *Autism in Adulthood*, *3*, 52–60.
Pellicano, E. (2020). Commentary: Broadening the research remit of participatory methods in autism science – a commentary on Happé and Frith (2020). *Journal of Child Psychology and Psychiatry*, *61*, 233–235.
Pellicano, E., & Burr, D. (2012). When the world becomes 'too real': A Bayesian explanation of autistic perception. *Trends in Cognitive Sciences*, *16*, 504–510.
Pellicano, E., Dinsmore, A. & Charman, T. (2013). A future made together: Shaping autism research in the UK. Available from: http://discovery.ucl.ac.uk/10017703/1/A_Future_Made_Together_1.2_LR.pdf. Accessed 20th February 2021
Pellicano, E., Dinsmore, A., & Charman, T. (2014a). Views on researcher-community engagement in autism research in the United Kingdom: A mixed-methods study. *PLoS One*, *9*, 1–11.
Pellicano, E., Dinsmore, A., & Charman, T. (2014b). What should autism research focus upon? Community views and priorities from the United Kingdom. *Autism*, *18*, 756–770.
Pellicano, E., Jeffery, L., Burr, D., & Rhodes, G. (2007). Abnormal adaptive face-coding mechanisms in children with autism spectrum disorder. *Current Biology*, *17*, 1508–1512.
Pellicano, E., & Stears, M. (2011). Bridging autism, science and society: Moving toward an ethically informed approach to autism research. *Autism Research*, *4*, 271–282.
Picardi, A., Gigantesco, A., Tarolla, E., Stoppioni, V., Cerbo, R., Cremonte, M., … & Nardocci, F. (2018). Parental burden and its correlates in families of children with autism spectrum disorder: A multicentre study with two comparison groups. *Clinical Practice & Epidemiology in Mental Health*, *14*, 143–176.
Pierson, P. (2000). Increasing returns, path dependence, and the study of politics. *American Political Science Review, 94*, 251–267.
Pripas-Kapit, S. (2020). Historicizing Jim Sinclair's "Don't Mourn for Us": A cultural and intellectual history of neurodiversity's first manifesto. In S.K. Kapp (Ed.), *Autistic community and the neurodiversity movement: Stories from the frontline* (pp. 23–39). Singapore: Palgrave Macmillan.

Raymaker, D.M. (2020). Shifting the system: AASPIRE and the loom of science and activism. In S.K. Kapp (Ed.), *Autistic community and the neurodiversity movement: Stories from the frontline* (pp. 133–145). Singapore: Palgrave Macmillan.
Raymaker, D.M., Teo, A.R., Steckler, N.A., Lentz, B., Scharer, M., Delos Santos, A., ... & Nicolaidis, C. (2020). "Having all of your internal resources exhausted beyond measure and being left with no clean-up crew": Defining autistic burnout. *Autism Adulthood*, *2*, 132–143.
Remington, A., & Fairnie, J. (2017). A sound advantage: Increased auditory capacity in autism. *Cognition*, *166*, 459–465.
Robertson, S.M. (2010). Neurodiversity, quality of life, and autistic adults: Shifting research and professional focuses onto real-life challenges. *Disability Studies Quarterly*, *30*. Available from: http://dsq-sds.org/article/view/1069/1234. Accessed 10th February 2021.
Roche, L., Adams, D., & Clark, M. (2021). Research priorities of the autism community: A systematic review of key stakeholder perspectives. *Autism*, *25*, 336–348.
Rottier, H., & Gernsbacher, M.A. (2020). Autistic adult and non-autistic parent advocates: Bridging the divide. In A. Carey, J. Ostrove & T. Fannon (Eds.), *Disability alliances and allies* (pp. 155–166). Bingley, UK: Emerald Publishing Limited.
Rutter, M. (2014). Addressing the issue of fractionation in autism spectrum disorder: A commentary on Brunsdon and Happé Frazier et al Hobson and Mandy et al *Autism*, *18*, 55–57.
Samson, F., Mottron, L., Soulieres, I., & Zeffiro, T.A. (2012). Enhanced visual functioning in autism: An ALE meta-analysis. *Human Brain Mapping*, *33*, 1553–1581.
Sasson, N.J., Faso, D.J., Nugent, J., Lovell, S., Kennedy, D.P., & Grossman, R.B. (2017). Neurotypical peers are less willing to interact with those with autism based on thin slice judgments. *Scientific Reports*, *7*, 40700.
Schroeder, J.H., Cappadocia, M.C., Bebko, J.M., Pepler, D.J., & Weiss, J.A. (2014). Shedding light on a pervasive problem: A review of research on bullying experiences among children with autism spectrum disorders. *Journal of Autism and Developmental Disorders*, *44*, 1520–1534.
Scott, M., Milbourn, B., Falkmer, M., Black, M., Bölte, S., Halladay, A., ... & Girdler, S. (2019). Factors impacting employment for people with autism spectrum disorder: A scoping review. *Autism*, *23*, 869–901.
Shah, A., & Frith, U. (1983). An islet of ability in autistic children: A research note. *Journal of Child Psychology and Psychiatry*, *24*, 613–620.
Shattuck, P.T., Steinberg, J., Yu, J., Wei, X., Cooper, B.P., Newman, L., & Roux, A.M. (2014). Disability identification and self-efficacy among college students on the autism spectrum. *Autism Research and Treatment*, *2014*, 924182.
Sheppard, E., Pillai, D., Wong, G.T., Ropar, D., & Mitchell, P. (2016). How easy is it to read the minds of people with autism spectrum disorder? *Journal of Autism and Developmental Disorders*, *46*, 1247–1254.
Silberman, S. (2015). *Neurotribes: The legacy of autism and how to think smarter about people who think differently*. Crows Nest, Australia: Allen & Unwin.
Silverman, C. (2011). *Understanding autism: Parents, doctors, and the history of a disorder*. Princeton, NJ: Princeton University Press.
Sinclair, J. (1993). Don't mourn for us. Available from: http://www.autreat.com/dont_mourn.html. Accessed 15th February 2021.
Sinclair, J. (2012). Why I dislike "person first" language. In The Autistic Self Advocacy Network (Ed.), *Loud hands: Autistic people, speaking* (pp. 223–224). Washington, DC: The Autistic Press.
Singer, J. (1998). *Odd people in: The birth of community amongst people on the "Autistic Spectrum"*. Sydney, NSW: Faculty of Humanities and Social Science, University of Technology, Sydney.
Singh, J., Illes, J., Lazzeroni, L., & Hallmayer, J. (2009). Trends in US autism research funding. *Journal of Autism and Developmental Disorders*, *39*, 788–795.
Sonuga-Barke, E.J.S. (2020). Editorial: 'People get ready': Are mental disorder diagnostics ripe for a Kuhnian revolution? *Journal of Child Psychology and Psychiatry*, *61*, 1–3.
Sutera, S., Pandey, J., Esser, E.L., Rosenthal, M.A., Wilson, L.B., Barton, M., ... & Fein, D. (2007). Predictors of optimal outcome in toddlers diagnosed with autism spectrum disorders. *Journal of Autism and Developmental Disorders*, *37*, 98–107.
Swain, J., French, S., & Cameron, C. (2003). *Controversial issues in a disabling society*. Buckingham, UK: Open University Press.
Tereshko, L., Ross, R.K., & Frazee, L. (2021). The effects of a procedure to decrease motor stereotypy on social interactions in a child with autism spectrum disorder. *Behavior Analysis in Practice*, *14*, 367–377.
Union of the Physically Impaired Against Segregation (1976). Fundamental principles of disability. Available from: https://disability-studies.leeds.ac.uk/wp-content/uploads/sites/40/library/UPIAS-fundamental-principles.pdf. Accessed 18th October 2021.
Verhoeff, B. (2015). Fundamental challenges for autism research: The science-practice gap, demarcating autism and the unsuccessful search for the neurobiological basis of autism. *Medicine, Health Care and Philosophy*, *18*, 443–447.
Verriden, A.L., & Roscoe, E.M. (2019). An evaluation of a punisher assessment for decreasing automatically reinforced problem behavior. *Journal of Applied Behavior Analysis*, *52*, 205–226.
Vismara, L.A., & Rogers, S.J. (2010). Behavioral treatments in autism spectrum disorder: What do we know? *Annual Review of Clinical Psychology*, *6*, 447–468.
Walker, N. (2012). Throw away the master's tools: Liberating ourselves from the pathology paradigm. In The Autistic Self Advocacy Network (Ed.), *Loud hands: Autistic people, speaking* (pp. 225–237). Washington, DC: The Autistic Press.
Walker, N. (2014). Neurodiversity: Some basic terms & definitions. Available from: https://neuroqueer.com/neurodiversity-terms-and-definitions/. Accessed 18th October 2021.
Walker, N., & Raymaker, D.M. (2020). Toward a neuroqueer future: An interview with Nick Walker. *Autism in Adulthood*, *3*, 5–10. https://doi.org/10.1089/aut.2020.29014.njw
Warner, G., Parr, J.R., & Cusack, J. (2019). Workshop report: Establishing priority research areas to improve the physical health and well-being of autistic adults and older people. *Autism in Adulthood*, *1*, 20–26.
Webster, M.A. (2011). Adaptation and visual coding. *Journal of Vision*, *11*, 3.
Wechsler, D. (2008). *WAIS-IV: Wechsler Adult Intelligence Scale*. San Antonio, TX: Pearson.
Wechsler, D. (2014). *WISC-V: Technical and interpretive manual*. Bloomington, MN: Pearson.
Wilkenfeld, D., & McCarthy, A. (2020). Ethical concerns with applied behavior analysis for autism spectrum "disorder". *Kennedy Institute of Ethics Journal*, *30*, 31–69.
Williams, D. (1996). *Autism: An inside-out approach*. London: Jessica Kingsley Publishers.
Williams, E., Gleeson, K., & Jones, B.E. (2019). How pupils on the autism spectrum make sense of themselves in the context of their experiences in a mainstream school setting: A qualitative metasynthesis. *Autism*, *23*, 8–28.
Witkin, H., Oltman, P., Raskin, E., & Karp, S. (1971). *A manual for the Embedded Figures Test*. Palo Alto, CA: Consulting Psychologists Press.

Raymaker, D.M. (2020). Shifting the system: AASPIRE and the loom of science and activism. In S.K. Kapp (Ed.), *Autistic community and the neurodiversity movement: Stories from the frontline* (pp. 133–145). Singapore: Palgrave Macmillan.
Raymaker, D.M., Teo, A.R., Steckler, N.A., Lentz, B., Scharer, M., Delos Santos, A., ... & Nicolaidis, C. (2020). "Having all of your internal resources exhausted beyond measure and being left with no clean-up crew": Defining autistic burnout. *Autism Adulthood*, *2*, 132–143.
Remington, A., & Fairnie, J. (2017). A sound advantage: Increased auditory capacity in autism. *Cognition*, *166*, 459–465.
Robertson, S.M. (2010). Neurodiversity, quality of life, and autistic adults: Shifting research and professional focuses onto real-life challenges. *Disability Studies Quarterly*, *30*. Available from: http://dsq-sds.org/article/view/1069/1234. Accessed 10th February 2021.
Roche, L., Adams, D., & Clark, M. (2021). Research priorities of the autism community: A systematic review of key stakeholder perspectives. *Autism*, *25*, 336–348.
Rottier, H., & Gernsbacher, M.A. (2020). Autistic adult and non-autistic parent advocates: Bridging the divide. In A. Carey, J. Ostrove & T. Fannon (Eds.), *Disability alliances and allies* (pp. 155–166). Bingley, UK: Emerald Publishing Limited.
Rutter, M. (2014). Addressing the issue of fractionation in autism spectrum disorder: A commentary on Brunsdon and Happé Frazier et al Hobson and Mandy et al *Autism*, *18*, 55–57.
Samson, F., Mottron, L., Soulieres, I., & Zeffiro, T.A. (2012). Enhanced visual functioning in autism: An ALE meta-analysis. *Human Brain Mapping*, *33*, 1553–1581.
Sasson, N.J., Faso, D.J., Nugent, J., Lovell, S., Kennedy, D.P., & Grossman, R.B. (2017). Neurotypical peers are less willing to interact with those with autism based on thin slice judgments. *Scientific Reports*, *7*, 40700.
Schroeder, J.H., Cappadocia, M.C., Bebko, J.M., Pepler, D.J., & Weiss, J.A. (2014). Shedding light on a pervasive problem: A review of research on bullying experiences among children with autism spectrum disorders. *Journal of Autism and Developmental Disorders*, *44*, 1520–1534.
Scott, M., Milbourn, B., Falkmer, M., Black, M., Bölte, S., Halladay, A., ... & Girdler, S. (2019). Factors impacting employment for people with autism spectrum disorder: A scoping review. *Autism*, *23*, 869–901.
Shah, A., & Frith, U. (1983). An islet of ability in autistic children: A research note. *Journal of Child Psychology and Psychiatry*, *24*, 613–620.
Shattuck, P.T., Steinberg, J., Yu, J., Wei, X., Cooper, B.P., Newman, L., & Roux, A.M. (2014). Disability identification and self-efficacy among college students on the autism spectrum. *Autism Research and Treatment*, *2014*, 924182.
Sheppard, E., Pillai, D., Wong, G.T., Ropar, D., & Mitchell, P. (2016). How easy is it to read the minds of people with autism spectrum disorder? *Journal of Autism and Developmental Disorders*, *46*, 1247–1254.
Silberman, S. (2015). *Neurotribes: The legacy of autism and how to think smarter about people who think differently*. Crows Nest, Australia: Allen & Unwin.
Silverman, C. (2011). *Understanding autism: Parents, doctors, and the history of a disorder*. Princeton, NJ: Princeton University Press.
Sinclair, J. (1993). Don't mourn for us. Available from: http://www.autreat.com/dont_mourn.html. Accessed 15th February 2021.
Sinclair, J. (2012). Why I dislike "person first" language. In The Autistic Self Advocacy Network (Ed.), *Loud hands: Autistic people, speaking* (pp. 223–224). Washington, DC: The Autistic Press.
Singer, J. (1998). *Odd people in: The birth of community amongst people on the "Autistic Spectrum"*. Sydney, NSW: Faculty of Humanities and Social Science, University of Technology, Sydney.
Singh, J., Illes, J., Lazzeroni, L., & Hallmayer, J. (2009). Trends in US autism research funding. *Journal of Autism and Developmental Disorders*, *39*, 788–795.
Sonuga-Barke, E.J.S. (2020). Editorial: 'People get ready': Are mental disorder diagnostics ripe for a Kuhnian revolution? *Journal of Child Psychology and Psychiatry*, *61*, 1–3.
Sutera, S., Pandey, J., Esser, E.L., Rosenthal, M.A., Wilson, L.B., Barton, M., ... & Fein, D. (2007). Predictors of optimal outcome in toddlers diagnosed with autism spectrum disorders. *Journal of Autism and Developmental Disorders*, *37*, 98–107.
Swain, J., French, S., & Cameron, C. (2003). *Controversial issues in a disabling society*. Buckingham, UK: Open University Press.
Tereshko, L., Ross, R.K., & Frazee, L. (2021). The effects of a procedure to decrease motor stereotypy on social interactions in a child with autism spectrum disorder. *Behavior Analysis in Practice*, *14*, 367–377.
Union of the Physically Impaired Against Segregation (1976). Fundamental principles of disability. Available from: https://disability-studies.leeds.ac.uk/wp-content/uploads/sites/40/library/UPIAS-fundamental-principles.pdf. Accessed 18th October 2021.
Verhoeff, B. (2015). Fundamental challenges for autism research: The science-practice gap, demarcating autism and the unsuccessful search for the neurobiological basis of autism. *Medicine, Health Care and Philosophy*, *18*, 443–447.
Verriden, A.L., & Roscoe, E.M. (2019). An evaluation of a punisher assessment for decreasing automatically reinforced problem behavior. *Journal of Applied Behavior Analysis*, *52*, 205–226.
Vismara, L.A., & Rogers, S.J. (2010). Behavioral treatments in autism spectrum disorder: What do we know? *Annual Review of Clinical Psychology*, *6*, 447–468.
Walker, N. (2012). Throw away the master's tools: Liberating ourselves from the pathology paradigm. In The Autistic Self Advocacy Network (Ed.), *Loud hands: Autistic people, speaking* (pp. 225–237). Washington, DC: The Autistic Press.
Walker, N. (2014). Neurodiversity: Some basic terms & definitions. Available from: https://neuroqueer.com/neurodiversity-terms-and-definitions/. Accessed 18th October 2021.
Walker, N., & Raymaker, D.M. (2020). Toward a neuroqueer future: An interview with Nick Walker. *Autism in Adulthood*, *3*, 5–10. https://doi.org/10.1089/aut.2020.29014.njw
Warner, G., Parr, J.R., & Cusack, J. (2019). Workshop report: Establishing priority research areas to improve the physical health and well-being of autistic adults and older people. *Autism in Adulthood*, *1*, 20–26.
Webster, M.A. (2011). Adaptation and visual coding. *Journal of Vision*, *11*, 3.
Wechsler, D. (2008). *WAIS-IV: Wechsler Adult Intelligence Scale*. San Antonio, TX: Pearson.
Wechsler, D. (2014). *WISC-V: Technical and interpretive manual*. Bloomington, MN: Pearson.
Wilkenfeld, D., & McCarthy, A. (2020). Ethical concerns with applied behavior analysis for autism spectrum "disorder". *Kennedy Institute of Ethics Journal*, *30*, 31–69.
Williams, D. (1996). *Autism: An inside-out approach*. London: Jessica Kingsley Publishers.
Williams, E., Gleeson, K., & Jones, B.E. (2019). How pupils on the autism spectrum make sense of themselves in the context of their experiences in a mainstream school setting: A qualitative metasynthesis. *Autism*, *23*, 8–28.
Witkin, H., Oltman, P., Raskin, E., & Karp, S. (1971). *A manual for the Embedded Figures Test*. Palo Alto, CA: Consulting Psychologists Press.

World Health Organisation (1980). International classification of impairments, disabilities, and handicaps. Available from: http://apps.who.int/iris/bitstream/handle/10665/41003/9241541261_eng.pdf;jsessionid=F5F67D582E80203398DD976AE4DB6F81?sequence=1. Accessed 13th February 2021

Yudell, M., Tabor, H.K., Dawson, G., Rossi, J., & Newschaffer, C., & Working Group in Autism Risk Communication and Ethics. (2013). Priorities for autism spectrum disorder risk communication and ethics. *Autism*, *17*, 701–722.

Accepted for publication: 6 September 2021

World Health Organisation (1980). International classification of impairments, disabilities, and handicaps. Available from: http://apps.who.int/iris/bitstream/handle/10665/41003/9241541261_eng.pdf;jsessionid=F5F67D582E80203398DD976AE4DB6F81?sequence=1. Accessed 13th February 2021

Yudell, M., Tabor, H.K., Dawson, G., Rossi, J., & Newschaffer, C., & Working Group in Autism Risk Communication and Ethics. (2013). Priorities for autism spectrum disorder risk communication and ethics. *Autism*, *17*, 701–722.

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